Measurement of Force Torque, Pressure, Temperature and Strain

Measurement of Force and Torque

Introduction
A force is defined as the reaction between two bodies. This reaction may be in the form of a tensile force (pull) or it may be a compressive force (push). Force is represented mathematically as a vector and has a point of application. Therefore the measurement of force involves the determination of its magnitude as well as its direction. The measurement of force may be done by any of the two methods.

i) Direct method: This involves a direct comparison with a known gravitational force on a standard mass example by a physical balance.

ii) Indirect method: This involves the measurement of the effect of force on a body. For example.

a) Measurement of acceleration of a body of known mass which is subjected to force.
b) Measurement of resultant effect (deformation) when the force is applied to an elastic member.

Direct method

A body of mass "m" in the earth's gravitational field experiences a force F which is given by

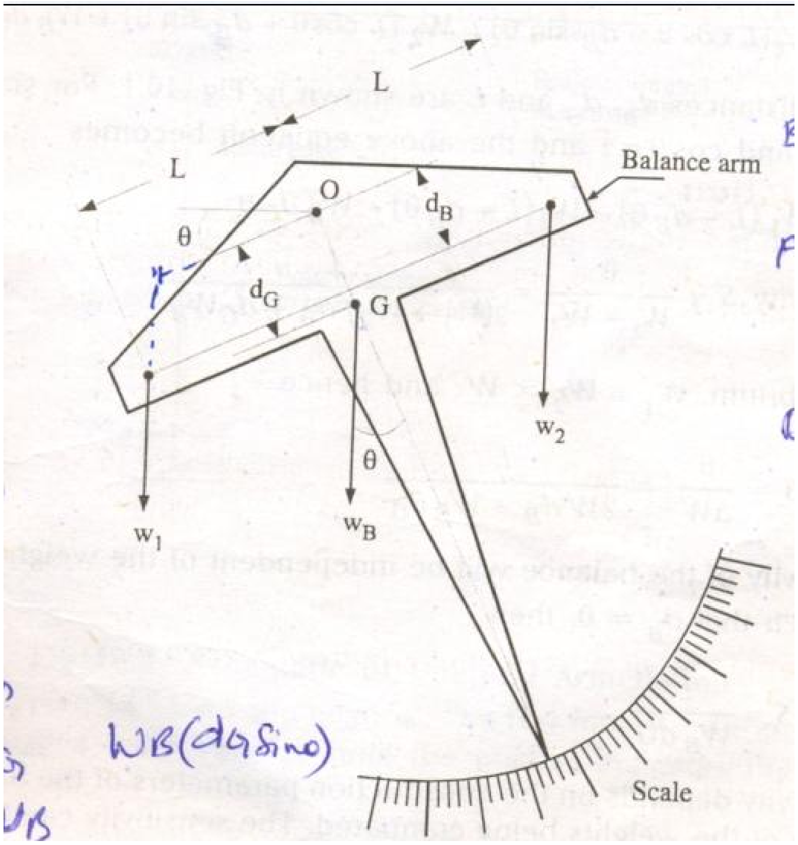
$$F = ma = W$$

Where ‘W’ is the weight of the body ‘a’ is the acceleration due to gravity. Any unknown force may be compared with the gravitational force (ma) on the standard mass ‘m’. The values of ‘m’ and ‘a’ should be known accurately in order to know the magnitude of the gravitation force.

Mass is a fundamental quantity and its standard kilogram is kept at France. The other masses can be compared with this standard with a precision of a few parts in109. On the other hand, ‘a’ is a derived quantity but still makes a convenient standard. Its value can be measured with an accuracy of 1 part in 106. Therefore any unknown force can be compared with the gravitational force with an accuracy of about this order of magnitude.

Analytical Balance :(Equal arm balance)

Direct comparison of an unknown force with the gravitational force can be explained with the help of an analytical balance. The direction of force is parallel to that of the gravitational force, and hence only its magnitude needs to be determined. The constructional details of an analytical balance is as shown in Fig.



The balance arm rotates about the point "O" and two forces W_1 and W_2 are applied at the ends of the arm. W_1 is an unknown force and W_2 is the known force due to a standard mass. Point G is the centre of gravity of the balance arm, and W_B is the weight of the balance arm and the pointer acting at G. The above figure show the balance is unbalanced position when the force W_1 and W_2 are unequal. This unbalance is indicated by the angle θ which the pointer makes with the vertical.

In the balanced position $W_1 = W_2$, and hence θ is zero. Therefore, the weight of the balance arm and the pointer do not influence the measurements.

The sensitivity S of the balance is defined as the angular deflection per unit of unbalance is between the two weight W_1 and W_2 and is given by

$$S = \frac{-}{w_1 - w_2} = \frac{-}{\Delta W}$$

where, ΔW is the difference between W_1 and W_2. The sensitivity S can be calculated by writing the moment equation at equilibrium as follows :

$$W_1 (L \cos\theta - d_B \sin\theta) = W_2 (L \cos\theta + d_B \sin\theta) + W_B\, d_G \sin\theta$$

where the distances d_B, d_G and L are shown in Fig. For small deflection angles sin θ = θ and cos θ = 1 and the above equation becomes

$$W_1 (L - d_B\,\theta) = W_2 (L + d_B\,\theta) + W_B\, d_G\,\theta$$

$$\therefore \textit{ The Sensitivity} \quad S = \frac{-}{w_1 - w_2} = \frac{L}{(w_1 + w_2) d_B + d_G W_B}$$

Near Equilibrium, $W_1 = W_2 = W$ and hence

$$S = \frac{-}{\Delta w} = \frac{L}{2W d_B + W_B\, d_G}$$

The sensitivity of the balance will be independent of the weight W Provided it is designed such that $d_B = 0$ then

$$S = \frac{L}{W_B\, d_G}$$

The sensitivity depends on the construction parameters of the balance arm and is independent of the weights being compared. The sensitivity can be improved by decreasing both d_G and W_B and increasing L. A compromise however, is to be struck between the sensitivity and stability of the balance.

Unequal Arm Balance

An equal arm analytical balance suffers from a major disadvantage. It requires a set of weights which are at least as heavy as the maximum weight to be measured. In order that the heavier weights may be measured with the help of lighter weights, balances with unequal arms are used.

The unequal arm balance uses two arms. One is called the **load arm** and the other is called the **power arm.** The load arm is associated with load i.e., the weight force to be measured, while power arm is associated with power i.e, the force produced by counter posing weights required to set the balance in equilibrium.

Fig. shows a typical unequal arm balance. Mass '*m'* acts as power on the beam and exerts a force of F_g due to gravity where F_g = m x g. This force acts as counterposing force against the load which may be a test force F_t.

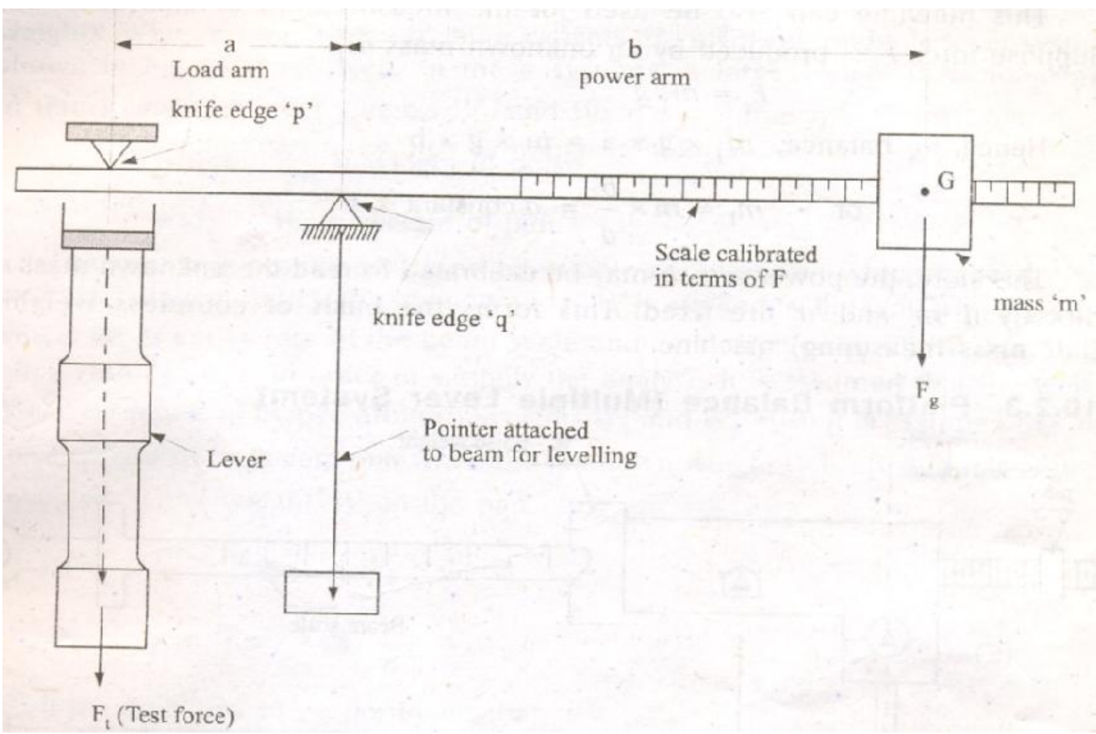


Fig. Schematic of Unequal Arm Balance

The beam is pivoted on a knife edge *'q'*. The test force F_t is applied by a screw or a lever through a knife edge *'p'* until the pointer indicates that the beam is horizontal.

For balance of moments, F_t x a = F_g x b
or test force F_t = F_g x b/a

$= m \times g \times b/a$
$= \text{constant} \times b$ (provided that g is constant)

Therefore the test force is proportional to the distance *'b'* of the mass from the pivot. Hence, if mass *'m'* is constant and the test force is applied at a fixed distance *'a'* from the knife edge *'q'* (i.e., the load arm is constant), the right hand of the beam (i.e., the power arm) may be calibrated in terms of force F_t. If the scale is used in different gravitational fields, a correction may be made for change in value of *'g'*.

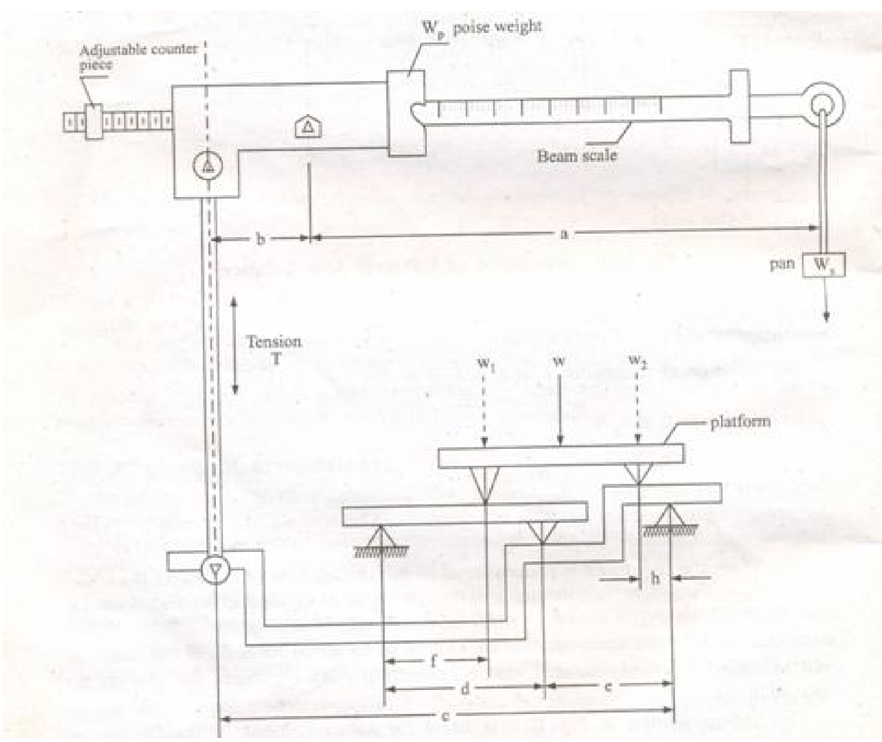
The set-up shown in Fig. is used for measurement of tensile force. With suitable modifications, it can be used for compression, shearing and bending forces.

This machine can also be used for the measurement of unknown mass. Suppose force F_t is Produced by an unknown mass m_t.

Therefore Ft = mt g
Hence, for balance, m1 x g x a = m x g x b
or m1 = m x b/a = a constant x b

Therefore, the power arm b may be calibrated to read the un known mass m1 directly if ‘m’ and ‘a’ are fixed. This forms the basis of countless weighing (i.e., mass measuring) machine

Platform Balance (Multiple Lever System)



Schematic of Multiple Lever System

An equal and unequal arm balances are not suited for measurement of large weights. When measurement of large weights are involved, multiple lever systems shown in Fig. are used. In these systems, a large weight W is measured in terms of two smaller weights Wp and Wg

where, Wp = weight of poise
and Ws = Weight of Pan

The system is provided with an adjustable counterpoise which is used to get an initial balance. Before the unknown load W is applied to the platform, the poise weight W_p is set at zero of the beam scale and counter piece is adjusted to obtain Initial zero balance.

In order to simplify the analysis it is assumed that the weight W can be replaced by two arbitrary weights W_1 and W_2. Also it is assumed that the poise weight W_p is at zero and when the unknown weight W is applied it is entirely balanced by the weight, W_S in the pan.

Therefore $T \times b = W_s \times a$ ….(1)

and $T \times c = W_1 f/d\ e + W_2 h$ …(2)

If the links are so proportioned that

$$h/e = f/d$$

We get : $T \times c = h (W_1 + W_2)\ hW$ ….(3)

From the above equation (3) it is clear that the weight W may be placed anywhere on the platform and its position relative to the two knife edges of the platform is immaterial.

T can be eliminated from equations. (l) and (3) to give

$$WS\,\frac{a}{b} = \frac{Wh}{d}$$

$$\text{Unknown weight } W = \frac{a}{b}\,\frac{c}{h}\,W_s$$

$$\text{where } m = \frac{a}{b}\,\frac{c}{h} \quad \text{is called themultiplication ration of the scale}$$

The multiplication ratio *M*, is indicative of weight that should be put in the pan to balance the weight on the platform. Suppose the scale has a multiplication ratio of 1000. It means that a weight of 1 kg put in the pan can balance a weight of 1000 kg put on the platform. Scales are available which have multiplication ratios as high as 10,000.

If the beam scale is so divided that a movement of poise weight Wp by 1 scale division represents a force of *x* kg, then a poise movement of y scale divisions should produce the same result as a weight W_p placed on the pan at the end of the beam. Hence, Wp y = x y a

$$or \quad x = \frac{W_p}{a}$$

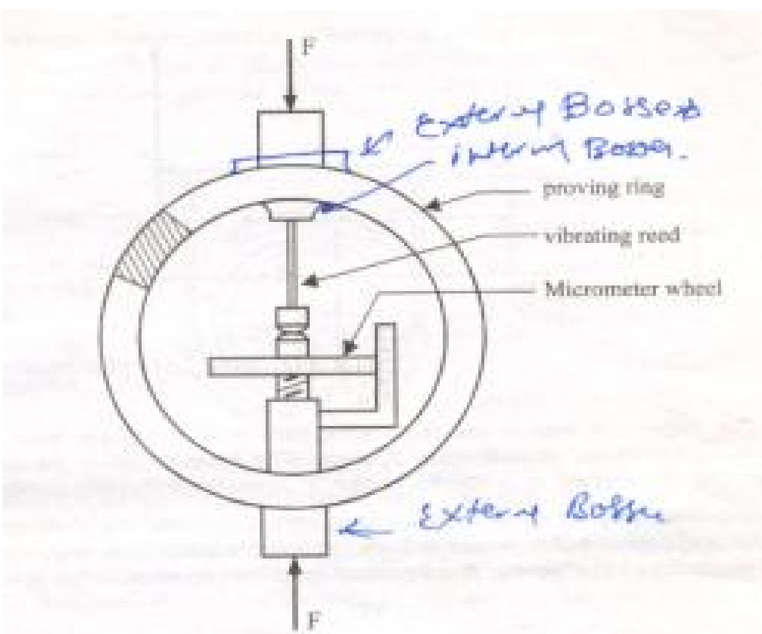
The above equation represents a relationship that determines the required scale divisions on the beam for any poise weight Wp

Proving Ring

This device has long been the standard for calibrating tensile testing machines and is in general, the means by which accurate measurement of large static loads may be obtained. A proving ring is a circular ring of rectangular cross section as shown in the Fig. which may be subjected to tensile or compressive forces across its diameter. The force-deflection relation for a thin ring is

$$F = \frac{16}{(\pi/2 - 4/\pi)} \frac{EI}{D^3} y$$

where, *F* is the force, E is the young's modulus, I is the moment of inertia of the section about the centroidal axis of bending section. *D* is the outside diagram of the ring, y is the deflection. The above equation is derived under the assumption that the thickness of the ring is small compared to the radius. And also it is clear that the displacement is directly proportional to the force.



The deflection is small and hence the usefulness of the proving ring as a calibration device depends on the accuracy with which this small deflection is measured. This is done by using a precision micrometer shown in the figure. In order to obtain precise measurements one edge of the micrometer is mounted on a vibrating reed device which is plucked to obtain a vibratory motion.

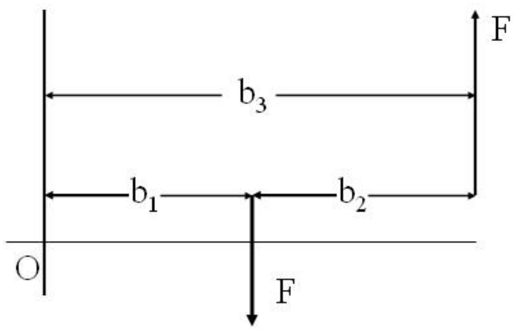
The micrometer contact is then moved forward until a noticeable damping of the vibration is observed.

Proving rings are normally used for force measurement within the range of 1.5 KN to 1.5 MN. The maximum deflection is typically of the order of 1% of the outside diameter of the ring.

Torque Measurement

The force, in addition to its effect along its line of action, may exert a turning effort relative to any axis other than those intersecting the line of action as shown in Fig. Such a turning effect is called torque or couple

Torque or couple $= Fb_1 - Fb_3$
$= Fb_2$



The important reason for measuring torque is to obtain load information necessary for stress or deflection analysis. The torque *T* may be computed by measuring the force *F* at a known radius *'r'* from the following relation $T=Fr$.

However, torque measurement is often associated with determination of mechanical power, either power required to operate a machine or power developed by the machine. The power is calculated from the relation.

$$P = 2 \pi NT$$

where *N* is the angular speed in revolutions per second. Torque measuring devices used in this connection are commonly known as **dynamometers.**

There are basically three types of dynamometers.

1. Absorption dynamometers: They absorb the mechanical energy as torque is measured, and hence are particularly useful for measuring power or torque developed by power sources such as engines or electric motors.

2. Driving dynamometers : These dynamometers measure power or torque and as well provide energy to operate the devices to be tested. They are, therefore, useful in determining performance characteristics of devices such as pumps, compressors etc

Transmission dynamometers : These are passive devices placed at an appropriate location within a machine or in between machines to sense the torque at that location. They neither add nor subtract the transmitted energy or power and are sometimes referred to as **torque meters.**

The first two types can be grouped as mechanical and electrical dynamometers.

Mechanical Dynamometer (Prony Brake)

These dynamometers are of absorption type. The most device is the prony brake as shown in Fig.

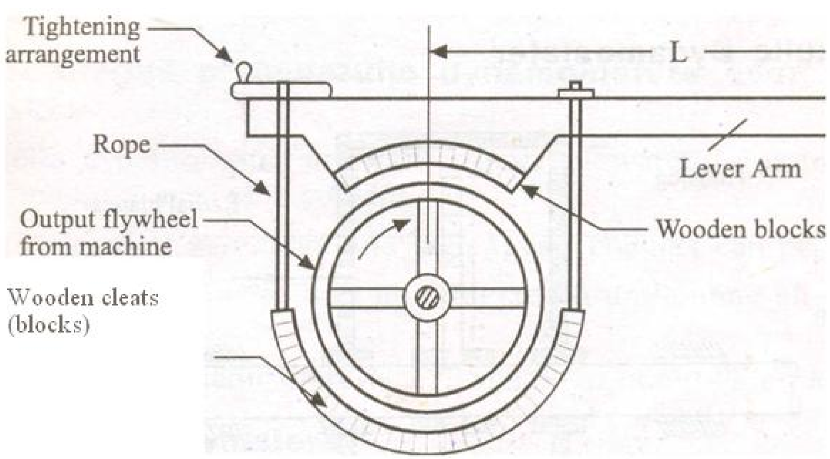


Fig. Schematic of Prony Brake

Two wooden blocks are mounted diametrically opposite on a flywheel attached to the rotating shaft whose power is to be measured. One block carries a lever arm, and an arrangement is provided to tighten the rope which is connected to the arm. The rope is tightened so as to increase the frictional resistance between the blocks and the flywheel. The torque exerted by the prony brake is.

$$T = F.L$$

where force F is measured by conventional force measuring instruments, like balances or load cells etc. The power dissipated in the brake is calculated by the following equation.

$$P = \frac{2\pi NT}{60} = \frac{2\pi FLN}{60} \ Watts.$$

where force F is in Newtons, L is the length of lever arm in meters, N is the angular speed in revolution per minute, and P in watts.

The prony brake is inexpensive, but it is difficult to adjust and maintain a specific load.

Limitation : The prony brake is inherently unstable. Its capacity is limited by the following factors.

i). Due to wear of the wooden blocks, the coefficient of friction varies between the blocks and the flywheel. This requires continuous tightening of clamp. Therefore, the system becomes unsuitable for measurement of large powers especially when used for long periods

ii) The use of prony brake results in excessive temperature rise which results in decrease in coefficient of friction leading to brake failure. In order to limit the temperature rise, cooling is required. This is done by running water into the hollow channel of the flywheel.

iii) When the machine torque is not constant, the measuring arrangement is subjected to oscillations. There may be changes in coefficient of friction and hence the reading of force F may be difficult to take.

Hydraulic Dynamometer

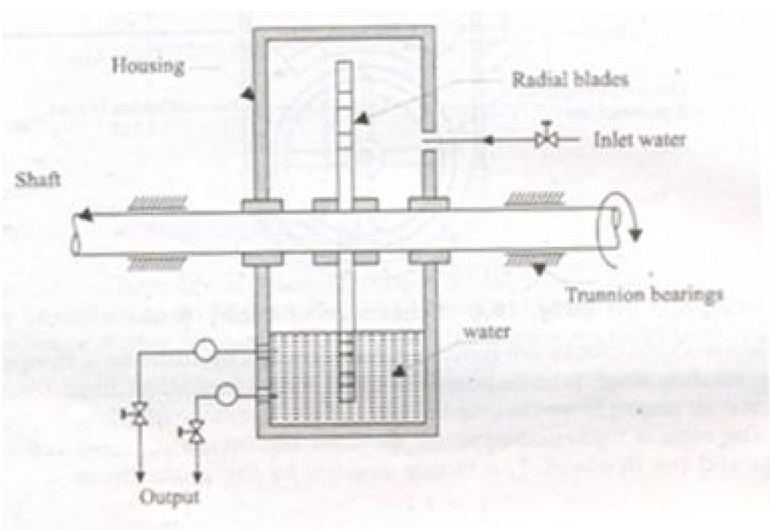


Fig. Section through a typical water brake

Fig. shows a hydraulic dynometer in its simplest form which acts as a water brake. This is a power sink which uses fluid friction for dissipation of the input energy and thereby measures the input torque-or power.

The capacity of hydraulic dynamometer is a function of two factors, speed and water level. The power consumed is a function of cube of the speed approximately. The torque is measured with the help of a reaction arm. The power absorption at a given speed may be controlled by adjustment of the water level in the housing. This type of dynamometer may be made in considerably larger capacities than the simple prony brake because the heat generated can be easily removed by circulating the water into and out of the housing. Trunnion bearings support the dynamometer housing, allowing it a freedom to rotate except for the restraint imposed by the reaction arm.

In this dynamometer the power absorbing element is the housing which tends to rotate with the input shaft of the driving machine. But, such rotation is constrained by a force-measuring device, such as some form of scales or load cell, placed at the end of a reaction arm of radius *r*. By measuring the force at the known radius, the torque *T* may be computed by the simple relation

Advantages of hydraulic dynamometers over mechanical brakes

- In hydraulic dynamometer constant supply of water running through the breaking medium acts as a coolant.
- The brake power of very large and high speed engines can be measured.
- The hydraulic dynamometer may be protected from hunting effects by means of a dashpot damper.
- In hydraulic dynamometer there is a flexibility in controlling the operation

Pressure Measurements

Introduction

Pressure is represented as a force per unit area exerted by a fluid on a container. The standard SI unit for pressure is Newton / Square meter (N/m2) or Pascal (Pa). High pressures can be conveniently expressed in KN/m2 while low pressure are expressed in terms of mm of water or mm of mercury.

Pressure is the action of one force against another over, a surface. The pressure P of a force F distributed over an area A is defined as:

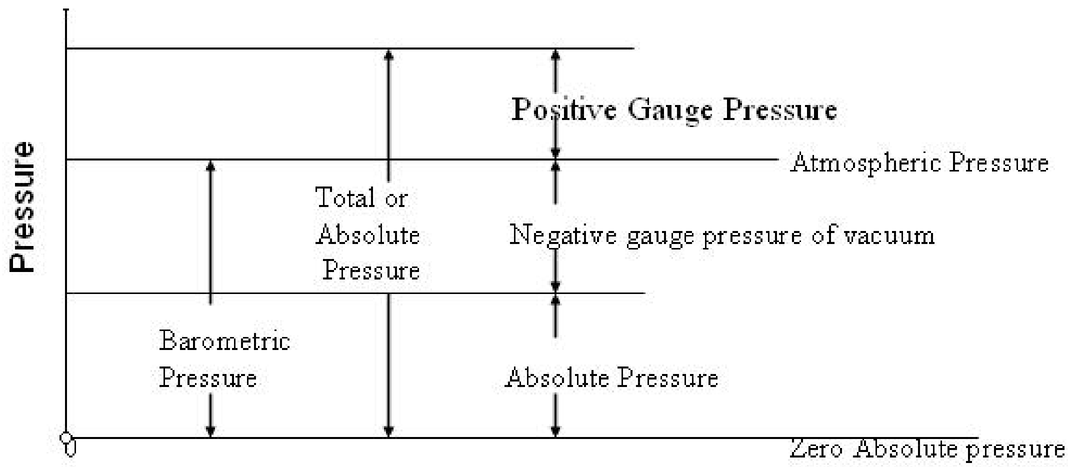
$$P = F/A$$

Units of Measure

System	Length	Force	Mass	Time	Pressure
MKS	Meter	Newton	Kg	Sec	N/M2 = Pascal
CGS	CM	Dyne	Gram	Sec	D/CM2
English	Inch	Pound	Slug	Sec	PSI

Absolute Pressure.

It refers to the absolute value of the force per unit are exerted on the containing wall by a fluid.



Relationship between Pressure Terms

Atmospheric Pressure

It is the pressure exerted by the earth's atmosphere and is usually measured by a barometer. At sea level. Its value is close to 1.013 x 105 N/m2 absolute and decreases with altitude

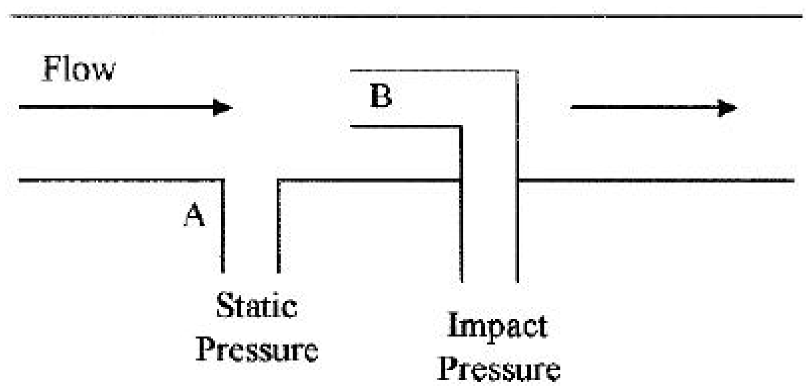
Gage Pressure

It represents the difference between the absolute pressure and the local atmosphere pressure
Vacuum
It is an absolute pressure less the atmospheric pressure i.e. a negative gage pressure.

Static and Dynamic pressures

If a fluid is in equilibrium, the pressure at a point is identical in all directions and independent of orientation is referred as pressure. In dynamic pressure, there exist a pressure gradient within the system. To restore equilibrium, the fluid flows from regions of higher pressure t regions of lower pressure.

Static, dynamic, and impact pressures



Static pressure is the pressure of fluids or gases that are stationary or not in motion. Dynamic pressure is the pressure exerted by a fluid or gas when it impacts on a surface or an object due to its motion or flow. In Fig., the dynamic pressure is (B − A).Impact pressure (total pressure) is the sum of the static and dynamic pressures on a surface or object. Point B in Fig. depicts the impact pressure.

Types of Pressure Measuring Devices

(i) Mechanical Instruments: These devices may be of two types. The first type includes those devices in which the pressure measurement is made by balancing an unknown pressure with a known force. The second types include those employing quantitative deformation of an elastic member for pressure measurements.

(ii) Electro-mechanical Instruments: this instrument employs a mechanical means for detecting the pressure and electrical means for indicating or recording the detected pressure.

(iii) Electronic Instruments: These instrument depends on some physical change which can be detected and indicated or recorded electronically.

Pressure- Measuring Transducer

Pressure is measured by transducing its effect to a deflection with the help of following types of transducers.

a) Gravitational types

(i) Liquid columns
(ii) Pistons or loose diaphragms, and weights.

b) Direct-acting elastic types

Unsymmetrically loaded tubes
Symmetrically loaded tubes
Elastic diaphragms
Bellows

Bulk compression

c) **Indirect-acting elastic type**

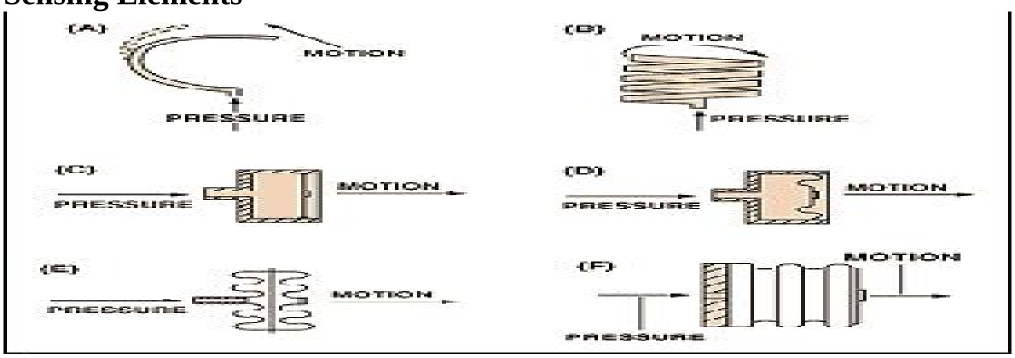
Piston with elastic restraining member

Use of Elastic Members in Pressure Measurement

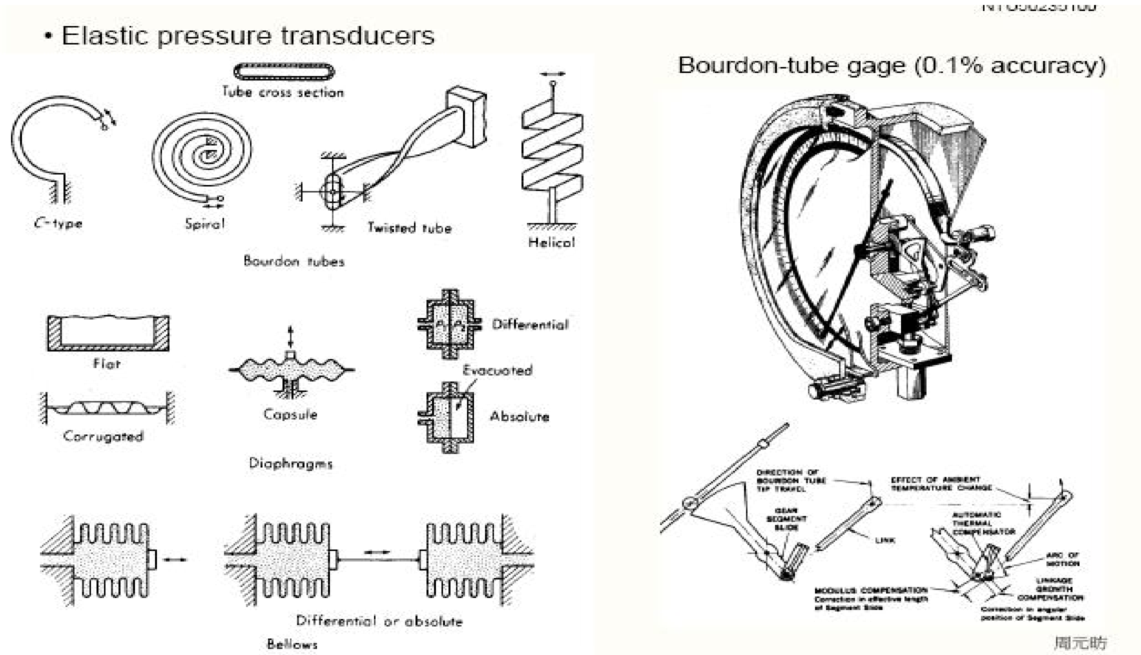
Application of pressure to certain materials causes elastic deformations. The magnitude of this elastic deformation can be related either analytically or experimentally to the applied pressure. Following are the three important elastic members used in the measurement of pressure.

(i) Bourdon tube,
(ii) Diaphragms and
(iii) Bellows

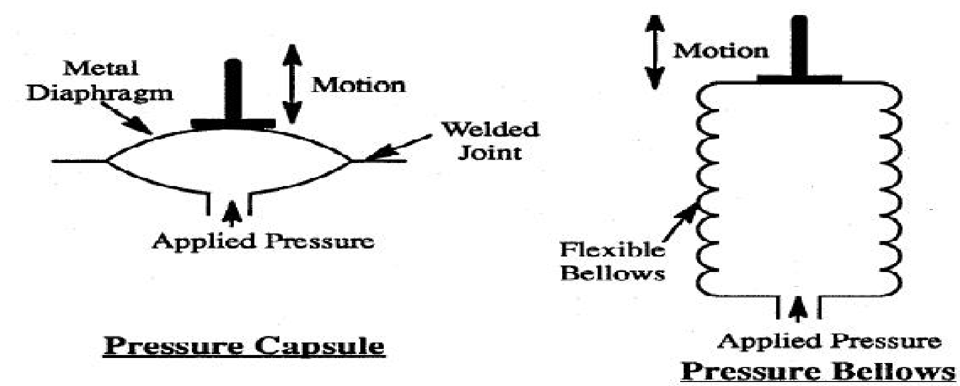
Sensing Elements

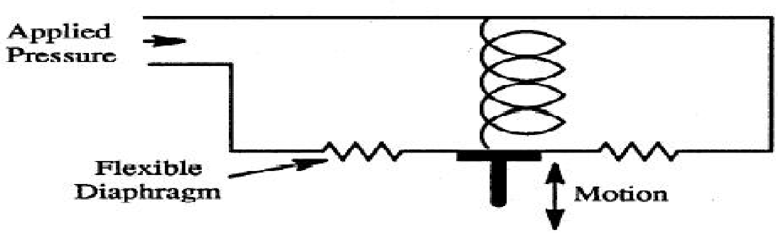


The basic pressure sensing element can be configured as a C-shaped Bourdon tube (A); a helical Bourdon tube (B); flat diaphragm (C); a convoluted diaphragm (D); a capsule (E); or a set of bellows (F).

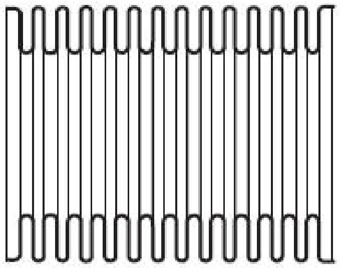


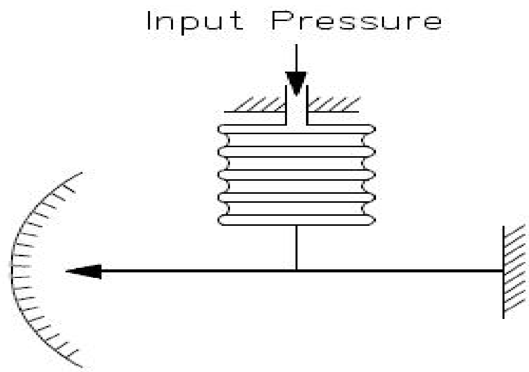
Primary Pressure Elements Capsule, Bellows & Spring Opposed Diaphragm



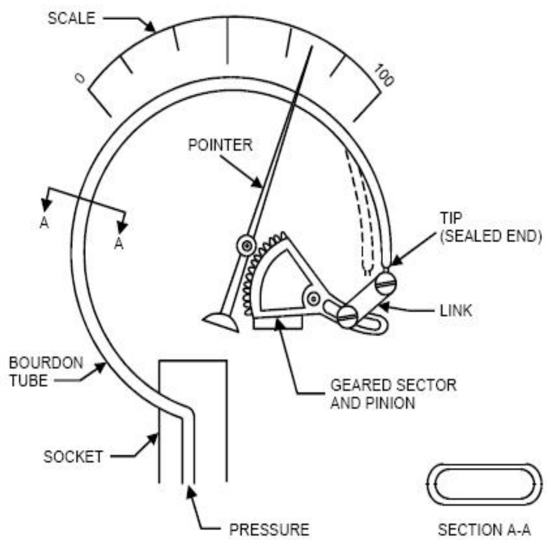


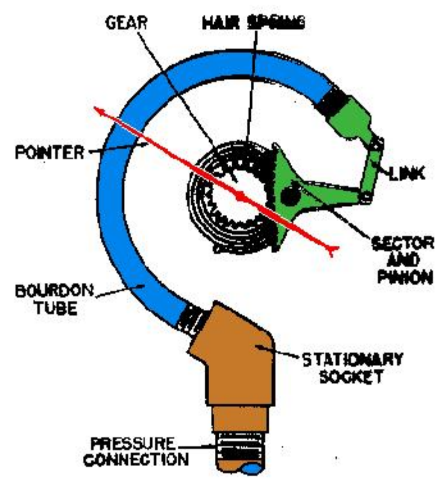
Bellows



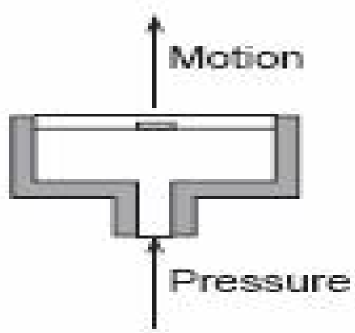


Bourdon Tube

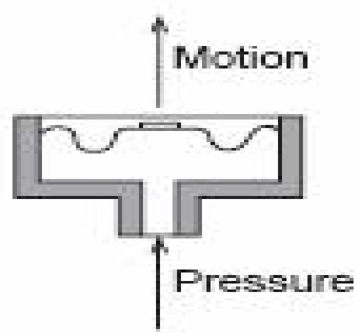




Diaphragm



(a)



(b)

A diaphragm usually is designed so that the deflection-versus-pressure characteristics are as linear as possible over a specified pressure range, and with a minimum of hysteresis and minimum shift in the zero point.

The Bridgman Gage

The resistance of fine wires changes with pressure according to the following linear relationship.

$$R = R1 (1 + \alpha p)$$

Where R1 Resistance at 1 atmosphere (100 KN/m2) in ohms

α Pressure coefficient of resistance in ohms/100 KN M-2

p gage pressure in KN/m2.

The above said resistance change may be used for measurement of pressures as high as 100,000 atm., 10.00KN/m2. A pressure transducer based on this principle is called a Bridgman gage. A typical gage uses a fine wire of manganin (84% Cu, 12% Mn, 4% Ni) wound in a coil and enclosed in a suitable pressure container. The pressure coefficient of resistance for this material is about 2.5x10-11 Pa-1. The total resistance of the wire is about 100Ω and conventional bridge circuits are employed for measuring the change in the resistance. Such gages are subjected to aging over a period of time, so that frequent calibration is required. However, when properly calibrated, the gage can be used for high pressure measurement with an accuracy of 0.1%. The transient response of the gage is exceedingly good. The resistance wire itself can respond of variations in the mega hertz range. Of course, the overall frequency response of the pressure-measurement system would be limited to much lower values because of the acoustic response of the transmitting fluid.

Low-Pressure measurement

In general, pressures below atmospheric may be called low pressures or vacuums. Its unit is micron, which is one-millionth of a meter (0.001 mm) of mercury column. Very low pressures may be defined as that pressures which are below 1 mm (1 torr) of mercury. An ultralow pressure is one which has pressure less than a millimicron(10-3 micron). An ultralow pressure is one which has pressure less than a millimicron (10-3 micron). Following are the two methods of measuring low pressure.

Direct Method : In this, direct measurement resulting in displacement caused by the action of pressure. Devices used in this method are Bourdon tubes, flat and corrugated-diaphragms, capsules and various forms of manometers. These devices are limited to a lowest pressure measurement of about 10mm of mercury.

Indirect or Inferential method : In this pressure is determined through the measurement of certain other pressure-controlled properties, such as volume, thermal conductivity etc.

The Mcleod Gage

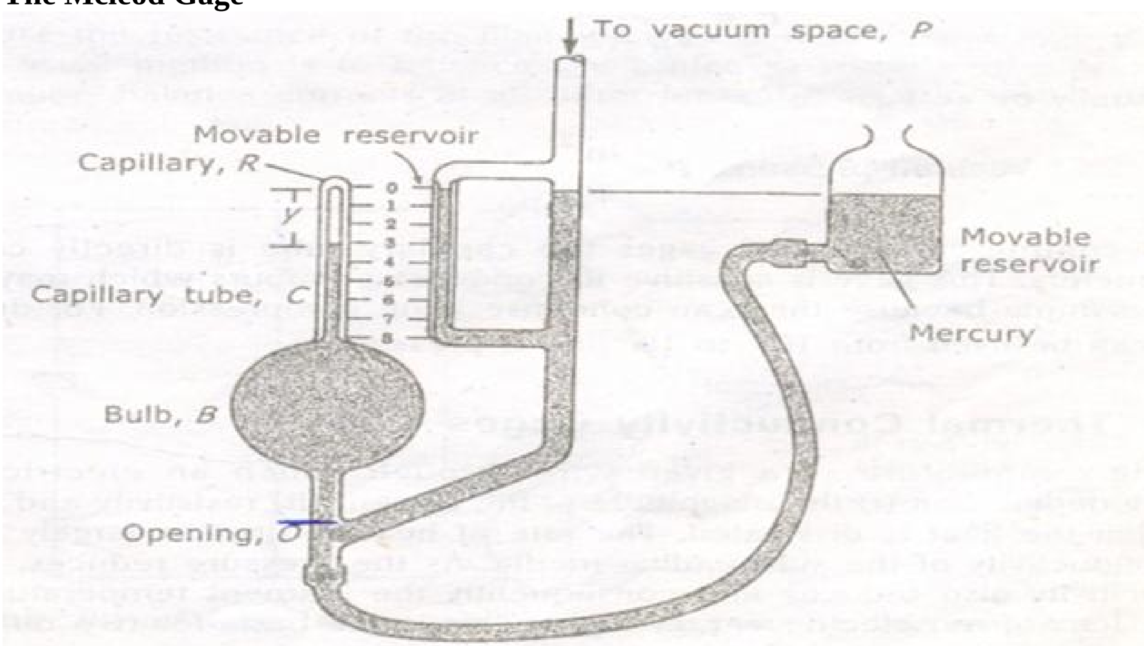
The operation of McLeod gage is based on Boyle's law.

$$p_1 = \frac{p_2\, v_2}{v_1}$$

Where, p1 andp2 are pressures at initial and conditions respectively, and v1 and v2 are volumes at the corresponding conditions. By compressing a known volume of low pressure gas to a higher pressure and measuring the resulting volume and pressure we can calculate the initial pressure.

The McLeod gage is a modified mercury manometer as shown in the Fig. 11.2. The movable reservoir is lowered until the mercury column drops below the opening O.

The Mcleod Gage



The Bulb B and capillary tube C are then at the same pressure as that of the vacuum pressure P. The reservoir is subsequently raised until the mercury fills the bulb and rises in the capillary tube to a point where the level in the reference capillary R is located at the zero point. If the volume of the capillary tube per unit length is 'a' then the volume of the gas in the capillary tube is.

Vc = ay ---------(1)

Where 'y' is the length of gas occupied in capillary tube.

If the volume of capillary tube, bulb and the tube down to the opening is VB. Assuming isothermal Compression, the pressure of the gas in the capillary tube is

$$P_C = P\frac{V_B}{V_C} \qquad \text{-------(2)}$$

The pressure indicated by the capillary tube is

$$Pc - P = \qquad \text{--------(3)}$$

Where, we are expressing the pressure in terms of the height of the mercury column. And combining equations (1), (2) and (3)

$$P = \frac{ay^2}{V_B - ay}$$

Usually ay << VB

$$\therefore \ Vacuumpressur, P = \frac{ay^2}{V_B}$$

In commercial McLeod gages the capillary tube is directly calibrated in micrometers. This gage is sensitive to condensed vapours which may be present in the sample because they can condense upon compression. For dry gases the gage can be used from 10^{-2} to 10^{2} µm of pressure.

Thermal Conductivity Gages

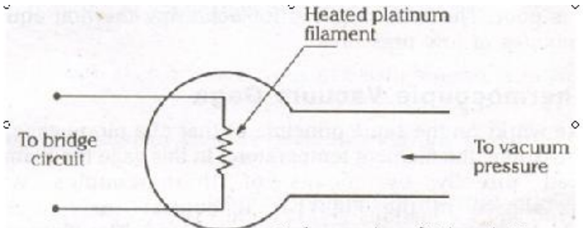
The temperature of a given wire through which an electric current is flowing depend, on (i) the magnitude of the current (ii) resistivity and (iii) the rate at which the heat is dissipated. The rate of heat dissipation largely depends on the conductivity of the surrounding media.

As the pressure reduces, the thermal conducitivity also reduces and consequently the filament temperature becomes higher for a given electric energy input. This is the basis for two different forms of gages to measure low pressures.

i). Pirani thermal conductivity gage
ii). Thermocouple vacuum gage

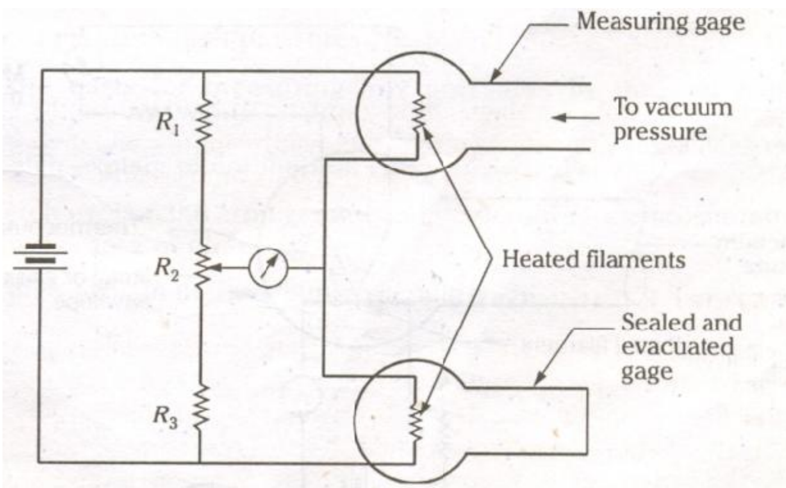
Pirani Thermal Conductivity Gages

The pirani gage as shown in the Fig. operates on the principle that if a heated wire is placed in a chamber of gas, the thermal conductivity of the gas depends on pressure. Therefore the transfer of energy from the wire to the gas is proportional to the gas pressure. If the supply of heating energy to the filament is kept constant and the pressure of the gas is varied, then the temperature of the filament will alter and is therefore a method of pressure measurement.



Schematic of Pirani Gage

To measure the resistance of the filament wire a resistance bridge circuit is used. The usual method is to balance the bridge at some datum pressure and use the out-of-balance currents at all other pressures as a measure of the relative pressures.



Pirani gage arrangement to compensate for ambient temperature Changes

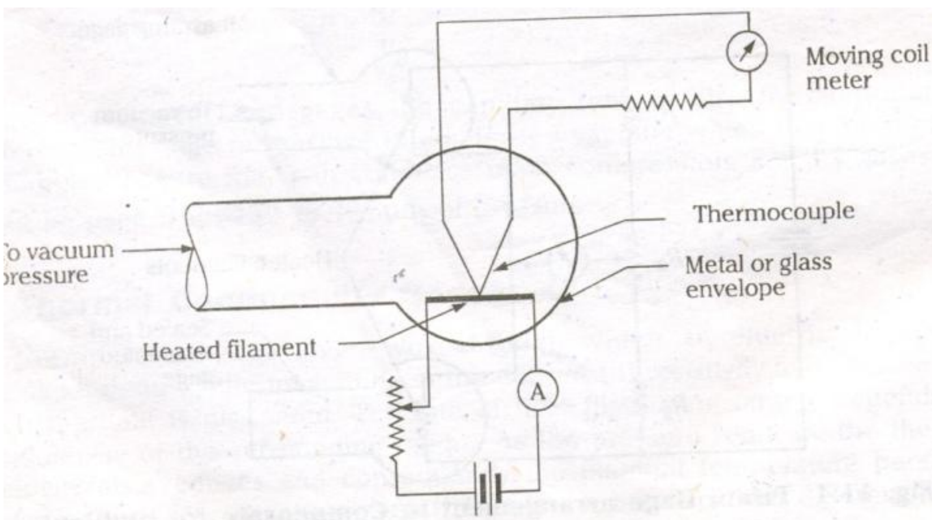
The heat loss from the filament is also a function of ambinet temperature and compensation for this effect may be achieved by connecting two gages in series as shown in Fig. The measuring gage is first evacuated and both the measuring and sealed gages are exposed to the same environment conditions. The bridge circuit is then adjusted through the resistor *R2* to get a null condition. When the measuring gage is exposed to the test vacuum pressure, the deflection of the bridge from the null position will be compensated for changes in environment temperature.

Pirani gages require calibration and are not suitable for use at pressures below l)..lm and upper limit is about 1 torr. For higher pressures, the thermal conductivity changes very little with pressure. It must be noted that the heat loss from the filament is also a function of the conduction losses to the filament supports and radiation losses to the surroundings. The transient response of the pirani gage is poor. The time required for achieving thermal equilibrium may be of several minutes at low pressures.

Thermocouple Vacuum gage

This gage works on the same principle as that of a pirani gage, but differ in the means for measuring the filament temperature. In this gage the filament temperature is measured directly by mc;ans of thermocouples welded directly to them as shown in the Fig. 11.5. It consists of heater filament and thermocouple enclosed in a glass or metal envelope.

The filament is heated by a constant current and its temperature depends upon the amount of heat lost to the surroundings by conduction and convection. At low pressures, the temperature of the filament is a function of the pressure of surrounding gas. Thus, the thermocouple provides an output voltage which is a function of temperature of the filament and consequently the pressure of the surrounding gas. The moving coil instrument may be directly calibrated to read the pressure.



Thermocouple Vacuum Gage

Temperature Measurements

Introduction

Temperature measurement is the most common and important measurement in controlling any process. Temperature may be defined as an indication of intensity of molecular kinetic energy within a system. It is a fundamental property similar to that of mass, length and time, and hence it is difficult to define. Temperature cannot be measured using basic standards through direct comparison. It can only be determined through some standardized calibrated device.

Change in temperature of a substance causes a variety of effects such as:

i) Change in physical state,
ii) Change in chemical state,
iii) Change in physical dimensions,
iv) Change in electrical properties and
v) Change in radiating ability.

And any of these effects may be used to measure the temperature

The change in physical and chemical states cannot be used for direct temperature measurement. However, temperature standards are based on changes in physical state. A change in physical dimension due to temperature shift forms the basis of operation for liquid-in-glass and bimetallic thermometers. Changes in electrical properties such as change in electrical conductivity and thermoelectric effects which produce electromotive force forms the basis for thermocouples. Another temperature-measuring method using the energy radiated from a hot body forms the basis of operation of optical radiation and infrared pyrometers.

Temperature Measurement by Electrical Effects

Electrical methods of temperature measurement are very convenient because they provide a signal that can be easily detected, amplified, or used for control purposes. In addition, they are quite accurate when properly calibrated and compensated. Several temperature-sensitive electrical elements are available for measuring temperature. Thermal emf and both positive and negative variations in resistance with temperature are important among them.

Thermo resistive Elements

The electrical resistance of most materials varies with temperature. Resistance elements which are sensitive to temperature are made of metals and are good conductors of electricity. Examples are nickel, copper, platinum and silver. Any temperature-measuring device which uses these elements are called resistance thermometers or resistance temperature detectors (RTD). If semiconducting materials like combination of metallic oxides of cobalt, manganese and nickel having large negative resistance co- efficient are used then such devices are called thermistors.

The differences between these two kinds of devices are :

Sl. No	Resistance Thermometer	Thermistor
1	In this resistance change with temperature shift is small and positive.	In this resistance change with temperature shift is relatively large and negative
2	Provides nearly a linear temperature-resistance relation	Non-linear temperature resistance relation.
3	Practical operating temperature range is -250 to 1000°C	Practical operating temperature range is -100 to 275°C.
4	More time-stable hence provide better reproducibility with low hysteresis	Not time-stable

Electrical Resistance Thermometers

The desirable properties of resistance-thermometer materials are:

i) The material should permit fabrication in convenient sizes.
ii) Its thermal coefficient of resistivity should be high and constant
iii) They must be corrosion-resistant and should not undergo phase changes with in the temperature ranges
iv) Provide reproducible and consistent results.

Electrical Resistance Thermometers

Unfortunately, there is no universally acceptable material and the selection of a particular material depends on the compromises.

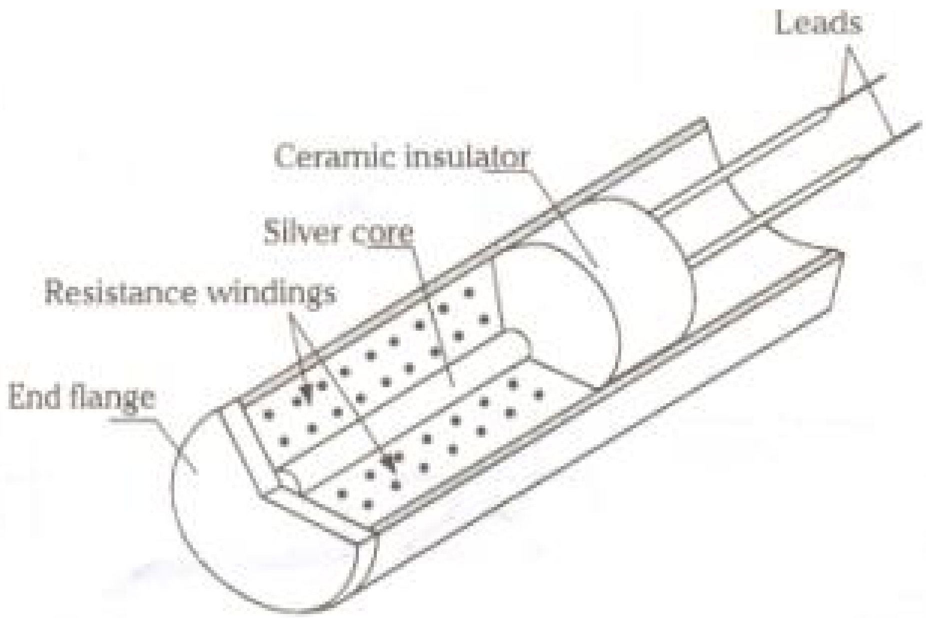
Although the actual resistance-temperature relation must be determined experimentally, for most metals the following empirical equation may be used.

$$R_t = Ro\ (l + aT + bT^2)$$

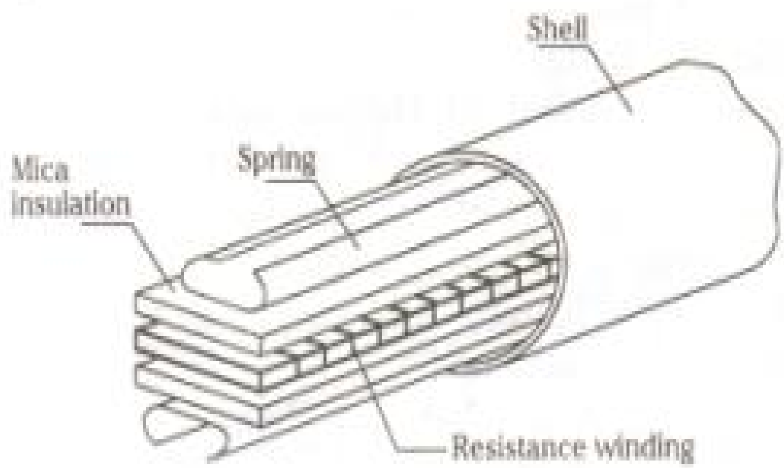
Where, *Rt* is the resistance at temperature T, *Ro* is the resistance at the reference temperature, *T* is the temperature and a and *b* are constants depending on the material.

Usually platinum, nickel and copper are the most commonly used materials, although others like tungsten, silver and iron can also be used.

Fig. shows the construction of two forms of resistance thermometer In Fig. (a) the element consists of a number of turns of resistance wire wrapped around a solid silver core. Heat is transmitted quickly from the end flange through the core to the windings.



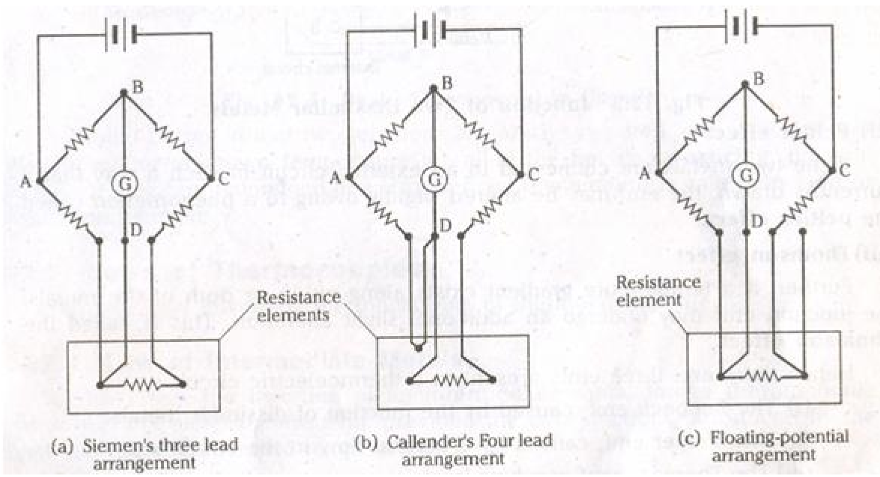
Another form of construction is shown in Fig. (b) in which the resistance wire is wrapped around a mica strip and sandwiched between two additional mica strips. These resistance thermometers may be used directly. But, when permanent installation with corrosion and mechanical protection is required a well or socket may be used.



Instrumentation for Resistance Thermometers

Some type of bridge circuit is normally used to measure resistance change in the thermometers. Leads of appropriate length are normally required, and any resistance change in them due to any cause affects the measurement. Hence, the lead resistance must be as low as possible relative to the element resistance

Three methods of compensating lead resistance error are as shown in the Fig. The arms *AD* and *DC* each contain the same length of leads. If the leads have identical properties and are at identical ambient conditions, then the effects introduced by one arm will be cancelled by the other arm.



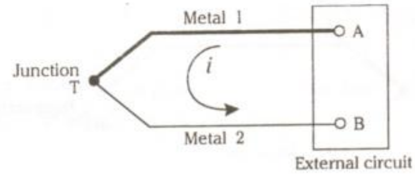
Methods of Compensating Lead Resistance Error

The Siemen's three-lead arrangement is the simplest corrective circuit. At balance conditions the centre lead carries no current, and the effect of the resistance of the other two leads is cancelled out. The Siemen's three-lead arrangement is the simplest corrective circuit. At balance conditions the centre lead carries no current, and the effect of the resistance of the other two leads is cancelled out. The calendar's four-lead arrangement solves the problem by inserting two additional lead wires in the adjustable leg of the bridge so that the effect of the lead wires on the resistance thermometer is cancelled out. The floating-potential arrangement is same as the Siemens' connection, but with an extra lead. This extra lead may be used to check the equality of lead resistance. The thermometer reading may be taken in the position shown, followed by additional readings with the two right and left leads interchanged, respectively. By averaging these readings, more accurate results may be obtained.

Usually, null-balance bridge is used but is limited to static or slowly changing temperatures. While the deflection bridge is used for rapidly changing temperatures.

1. See beck Effect:

When two dissimilar metals are joined together as shown in the Fig. an electromotive force (emf) will exists between the two points A and B, which is primarily a function of the junction temperature. This phenomenon is called the **see beck effect.**



Junction of Two Dissimilar Metals

ii) **Peltier effect**

If the two metals are connected to an external circuit in such a way that a current is drawn, the emf may be altered slightly owing to a phenomenon called the **peltier effect.**

iii) **Thomson effect**

Further, if a temperature gradient exists along either or both of the metals, the junction emf may undergo an additional slight alteration. This is called the **Thomson effect**.

Hence there are, three emfs present in a thermoelectric circuit:

i) The Seebeck emf, caused by the junction of dissimlar metals
ii) The Peltier emf, caused by a current flow in the circuit and
iii) The Thomson emf, resulting from a temperature gradient in the metals.

The Seebeck emf is important since it depends on the junction temperature.

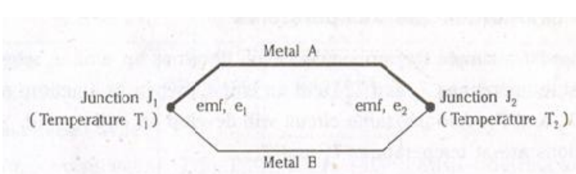
If the emf generated at the junction of two dissimilar metals is carefully measured as a function of temperature, then such a junction may be used for the measurement of temperature.

The above effects forms the basis for a thermocouple which is a temperature measuring element.

Thermocouple

If two dissimilar metals are joined an emf exists which is a function of several factors including the temperature.When junctions of this type are used to measure temperature, they are called thermocouples

The principle of a thermocouple is that if two dissimilar metals *A* and *B are* joined to form a circuit as shown in the Fig. It is found that when the two junctions J_1 and J_2 are at two different temperatures T_1 and T_2, small emf's e_1 and e_2 are generated at the junctions. The resultant of the two emf's causes a current to flow in the circuit. If the temperatures T_1 and T_2 are equal, the two emf's will be equal but opposed, and no current will flow. The net emf is a function of the two materials used to form the circuit and the temperatures of the two junctions. The actual relations, however, are empirical and the temperature-emf data must be based on experiment. It is important that the results are reproducible and therefore provide a reliable method for measuring temperature.



Basic Thermocouple Circuit

It should be noted that two junctions are always required, one which senses the desired or unknown temperature is called the **hot** or **measuring** junction. The other junction maintained at a known fixed temperature is called the **cold** or **reference** junction.

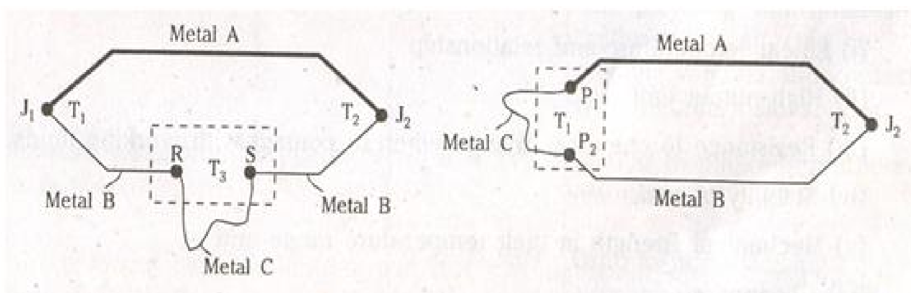
Laws of Thermocouples

The two laws governing the functioning of thermocouples are:

i) Law of Intermediate Metals:

It states that the insertion of an intermediate metal into a thermocouple circuit will not affect the net emf, provided the two junctions introduced by the third metal are at identical temperatures.

Application of this law is as shown in Fig. In Fig. (a), if the third metal C is introduced and the new junctions R and S are held at temperature T3, the net emf of the circuit will remain unchanged. This permits the insertion of a measuring device or circuit without affecting the temperature measurement of the thermocouple circuit

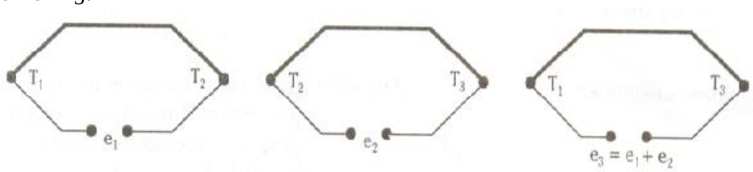


Circuits illustrating the Law of Intermediate Metals

In the Fig. (b) the third metal is introduced at either a measuring or reference junction. As long as junctions P_1 and P_2 are maintained at the same temperature T_P the net emf of the circuit will not be altered. This permits the use of joining metals, such as solder used in fabricating the thermocouples. In addition, the thermocouple may be embedded directly into the surface or interior of a conductor without affecting the thermocouple's functioning.

i) Law of Intermediate Temperatures:

It states that "If a simple thermocouple circuit develops an emf, e_1 when its junctions are at temperatures T_1 and T_2, and an emf *e_2, when its junctions are at temperature T_2 and T_3.* And the same circuit will develop an emf $e_3 = e_1 + e_2$, when its junctions are at temperatures T_1 and T_3.



Circuits illustrating the Law of Intermediate Temperatures

This is illustrated schematically in the above Fig. This law permits the thermocouple calibration for a given temperature to be used with any other reference temperature through the use of a suitable correction. Also, the extension wires having the same thermo-electric characteristics as those of the thermocouple wires can be introduced in the circuit without affecting the net emf of the thermocouple.

Thermocouple materials and Construction

Any two dissimilar metals can be used to form thermocouple, but certain metals and combinations are better than others. The desirable properties of thermocouple materials are:

i) Linear temperature-emf relationship
ii) High output emf,
iii) Resistance to chemical change when in contact with working fluids,
iv) Stability of emf,
v) Mechanical strength in their temperature range and
vi) Cheapness.

The thermocouple materials can be divided into two types

1. Rare-metal types using platinum, rhodium, iridium etc and
2. Base-metal types as given in the table.

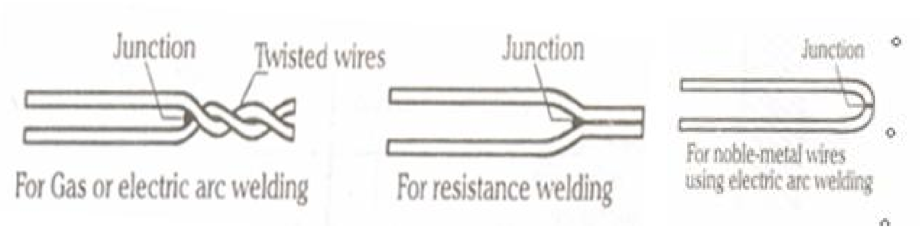
Table Thermocouple Ranges and Characteristics

Type	**ANSI Standards**	**Temperature Range °C**	**Characteristics**
A. Base-metal type			
1. Copper-constantan (40% Ni 60%Cu)	Type – T	-250 to + 400	Resists oxidizing and reducing atmospheres, and requires protection from acid fumes.
2. Iron Constantan	-	-200 to + 850	Low cost, corrodes in the presence of moisture, oxygen and sulphur bearing gases, suitable for reducing atmospheres
3. Chromel (90% Cr, 10% Ni) - Alumel (94% Ni, 2%Al + Si and Mn)	Type K	-200 to + 1100	Resistant to oxidizing but not to reducing atmospheres. Susceptible to attack by carbon-bearing gases, sulphur and cyanide fumes.
4. Chromel –Constantan	Type – E	-200 to + 850	Similar to Chromel Alumel

A. Rare-metal type 1) Platinum 90% rhodium 10% - Platinum	Type – S	0 to + 1400	Law emf, good resistance to oxidizing atmospheres, poor with reducing atmospheres. Calibration is affected by metallic vapours and contact with metallic oxides
2. Rhodium iridium – Iridium	Type - R	0 to + 2100	Similar to Platinum / rhodium - platinum
3. Tungsten (95%) rhenium (5%) Tungsten (72%) rhenium (26%)	–	0 to + 2600	Used in non-oxidising atmospheres only. The 5% rhenium arm is brittle at room temperature
4. Platinum, rhodium (30%) Platinum, rhodium (6%)	Type - B	+ 850 to + 1800	Not for reducing atmosphere or vacuum. Generates high emf per degree.

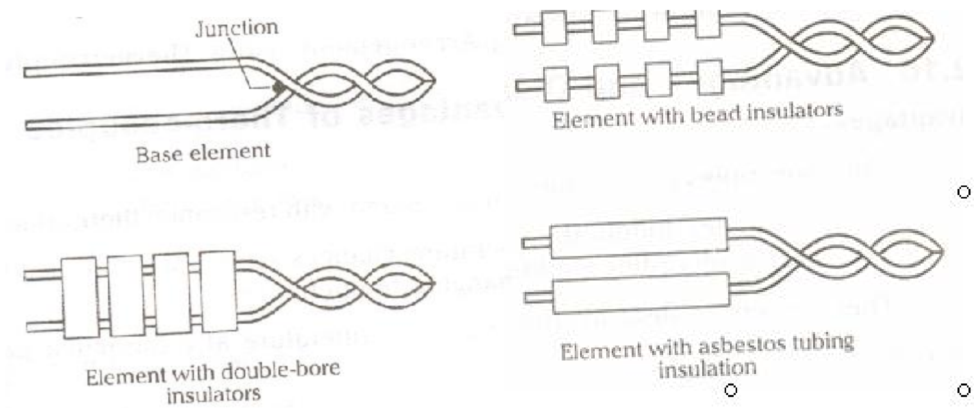
The size of thermocouple wire is important because higher the temperature to be measured, heavier should be the wire. As the wire size increases, the time response of the thermocouple to temperature change increases. Hence some compromise between time response and life of the thermocouples is required.

Thermocouples may be prepared by twisting the two wires together and brazed or welded as shown in the Fig.



Forms of Thermocouple Construction

Bare elements without any protector can be used for low-temperature thermocouples. But some form of protection is required for higher temperatures Fig. shows the common methods of providing insulation to the thermocouple wires.



Methods of Insulating Thermocouple Leads

Measurement of Thermal emf

The magnitude of emf developed by the thermocouples is very small (0.01 to 0.07 millivoIts/^{0}C), thus requires a sensitive devices to measure. Measurement of thermocouple output may be obtained by various ways. like millivolt meter or voltage-balancing potentiometer etc. Fig. shows a simple temperature-measuring system using a thermocouple as the sensing element and a potentiometer for indication. The thermoelectric circuit consists of a measuring junction J_1 and reference junction J_2, at the potentiometer. By the law of intermediate metals the potentiometer box may be considered to be an intermediate conductor. Assuming the two potentiometer terminals to be at identical temperature, the reference junction can be formed by the ends of the two thermocouple leads as they attach to the terminals. The reference temperature is determined using liquid-in-glass thermometer placed near the terminals. The value of the emf developed by the thermocouple circuit is measured using the potentiometer. Then using the table (values of emf Vs temperature) the temperature of the measuring junction can be determined.

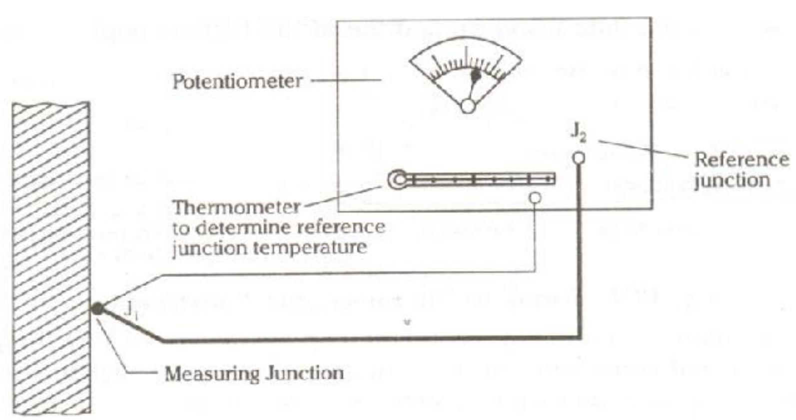


Fig. Temperature measuring Arrangement using Thermocouple

Advantages and Disadvantages of Thermocouples

Advantages

1. Thermocouples are cheaper than the resistance thermometers.
2. Thermocouples follow the temperature changes with small time lag thus suitable for recording rapidly changing temperatures.
3. They are convenient for measuring the temperature at a particular point.

Disadvantages

1. Possibility of inaccuracy due to changes in the reference junction temperature hence they cannot be used in precision work.

2. For long life, they should be protected to prevent contamination and have to be chemically inert and vacuum tight.

3. When thermocouples are placed far from the measuring systems, connections are made by extension wires. Maximum accuracy is obtained only when compensating wires are of the same material as that of thermocouple wires, thus the circuit becomes complex.

Theory of Radiation Pyrometers

When temperatures to be measured are very high and physical contact with the medium to be measured is impossible, then thermal radiation methods or optical pyrometers are used. These pyrometers are used when corrosive vapours or liquids would destroy thermocouples and resistance thermometers which come in contact with the measuring medium .

Radiation pyrometers measures the heat emitted or reflected by a hot object. Thermal radiation is the electromagnetic radiation emitted as a result of temperature. The operation of thermal radiation pyrometers are based on black body concept. The total thermal radiation emitted by a black body per unit area is given by Stefen - Boltzmann law as.

$q_b = \sigma T^4$ W/m2

where, σ is the Stefen-Boltzmann constant = 56.7×10^{-9} W/m^2-k^4, and T is the absolute temperature of the surface in kelvin.

Prevost's theory of exchange states that, for two block bodies in sight, each will radiate energy to the other and hence the net energy transfer per unit area from one to the other is given by.

$$q_b = \sigma \left(T_1^4 - T_2^4\right) \quad \mathrm{W/m^2}$$

Where, T_1 and T_2 are the absolute surface temperatures and $T_1 > T_2$. If T_1 is much higher than T_2 it can be assumed that the radiation is proportional toT_1^4 as the term T_2^4 becomes insignificant.

A rough black surface radiates more heat than a smooth bright surface. This effect is called emissivity and is expressed as.

€ = q / qb

The energy is radiated over a range of frequencies of the electromagnetic spectrum, the distribution for any particular wavelength (λ) is given by plank's radiation law as,

$$qb^{\text{a}} = C_1 / \{\text{a}^5 \left(eC^{C2/\text{a}T} - 1\right)\}$$

where, *qbλ* is the energy radiated at wavelength λ and C1 and C2 are constants.

Energy distribution curves calculated from this equation are shown in Fig. for three temperature values and the small visible band of the range is indicated. If a vertical line is drawn at a particular frequency value, then the radiated energy has a particular intensity at each temperature. If a vertical band representing a range of frequencies is drawn, the energy radiated at a particular temperature is given by the area in the band under that temperature curve. The values given in the figure are for perfect black bodies. These values should be multiplied by emissivity in order to get values for actual surfaces. Actual surfaces exhibit highly variable emissivities over the wavelength spectrum. And for the purpose of analysis, the actual surfaces are approximated as grey bodies

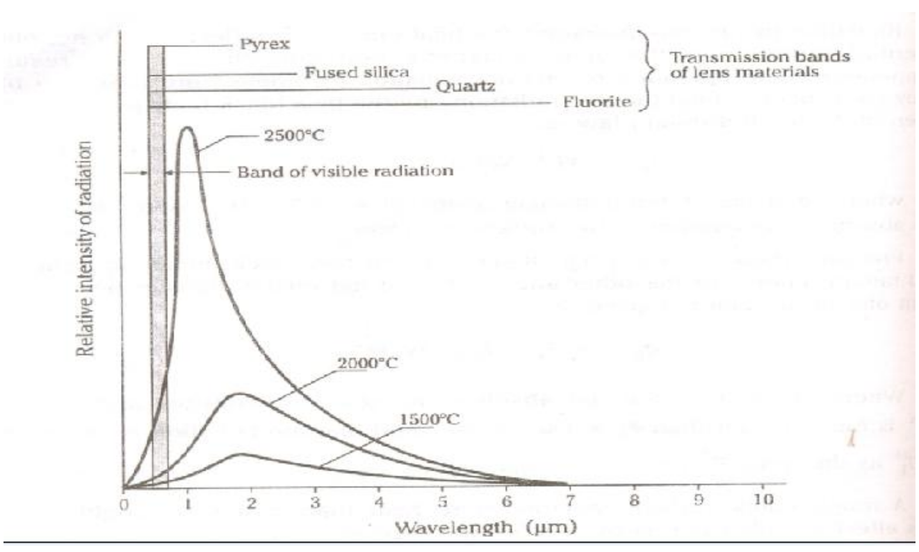


Fig. Distribution of Radient Energy for Black Bodies

It is apparent that the intensity of radiation varies appreciably with wavelength. Also it is observed that the point of maximum radiant intensity shifts to the shorter wavelengths as the temperature increases. This is a common phenomenon observed in the change of colour of a body being heated. A metal gradually heated changes its colour from red, which has a long wavelength to yellow and white as the intensity of radiation increases at the shorter wavelengths of the visible spectrum.

Principles used for Radiation Temperature Measuring Devices

1. Total Radiation Pyrometry:
In this case the total radiant energy from a heated body is measured. This energy is represented by the area under the curves of above Fig. and is given by Stefen - Boltzmann law. The radiation pyrometer is intended to receive maximum amount of radiant energy at wide range of wavelengths possible.

2.Selective Radiation Pyrometry:
This involves the measurement of spectral radiant intensity of the radiated energy from a heated body at a given wavelength. For example, if a vertical line is drawn in Fig. the variation of intensity with temperature for given wavelength can be found. The optical pyrometer uses this principle.

Total Radiation Pyrometers

The total radiation pyrometers receives all the radiations from a hot body and focuses it on to a sensitive temperature transducer like thermocouple, resistance thermometer etc. It consists of a radiation-receiving element and a measuring device to indicate the temperature. The most common type is shown in the Fig. A lens is used to concentrate the total radiant energy from the source on to the temperature sensing element. The diaphragms are used to prevent reflections. When lenses are used, the transmissibility of the glass determines the range of frequencies passing through. The transmission bands of some of the lens materials are shown in the Fig. The radiated energy absorbed by the receiver causes a rise of temperature. A balance is established between the energy absorbed by the receiver and that dissipated to the surroundings. Then the receiver equilibrium temperature becomes the measure of source temperature, with the scale established by calibration.

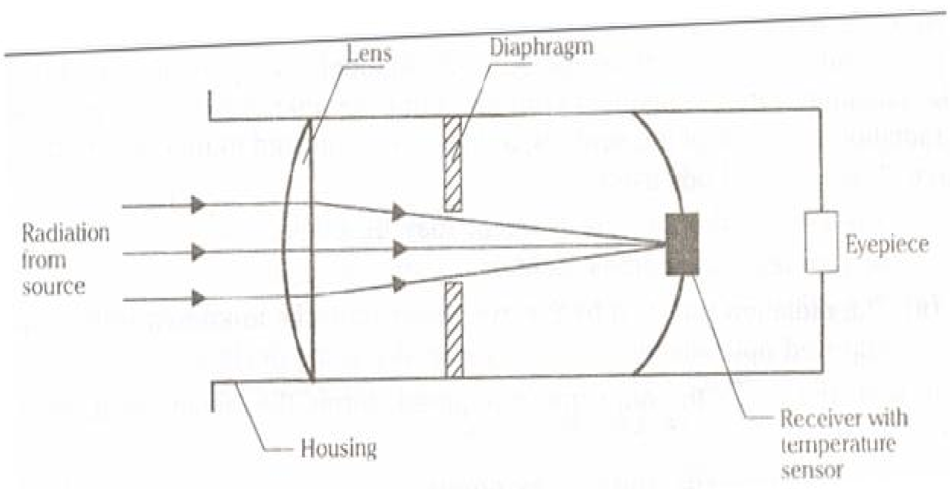


Fig. Schematic of Lens Type Radiation Receiving Device

The mirror type radiation receiver is another type of radiation pyrometer as shown in the Fig. Here the diaphragm unit along with a mirror is used to focus the radiation onto a receiver. The distance between the mirror and the receiver may be adjusted for proper focus. Since there is no lens, the mirror arrangement has an advantage a absorption and reflection effects are absent.

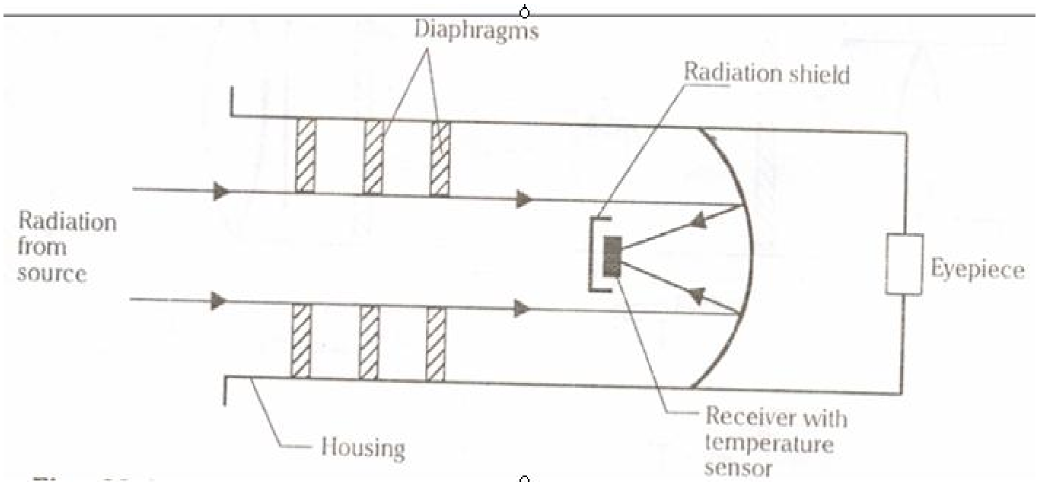


Fig. Mirror Focussing Type Radiation Receiving Device

Although radiation pyrometers may theoretically be used at any reasonable distance from a temperature source, there are practical limitations.

i) The size of target will largely determine the degree of temperature averaging, and in general, the greater the distance from the source, the greater the averaging.

ii) The nature of the intervening atmosphere will have a decided effect on the pyrometer indication. If smoke, dust or certain gases present considerable energy absorption may occur. This will have a particular problem when such absorbents are not constant, but varying with time. For these reasons, minimum practical distance is recommended

Optical pyrometers

Optical pyrometers use a method of matching as the basis for their operation. A reference temperature is provided in the form of an electrically heated lamp filament, and a measure of temperature is obtained by optically comparing the visual radiation from the filament with that from the unknown own source. In principle, the radiation from one of the sources, as viewed is adjusted to match with that from the other source. The two methods used are :

i) The current through the filament may be controlled electrically with the help of resistance adjustment or

ii) The radiation received by the pyrometer from the unknown source may be adjusted optically by means of some absorbing devices.

In both the cases the adjustment required, forms the means of temperature measurement.

The variable intensity optical pyrometer is, as shown in the Fig. The pyrometer is positioned towards an unknown temperature such that the objective lens focuses the source· in the plane of the lamp filament.

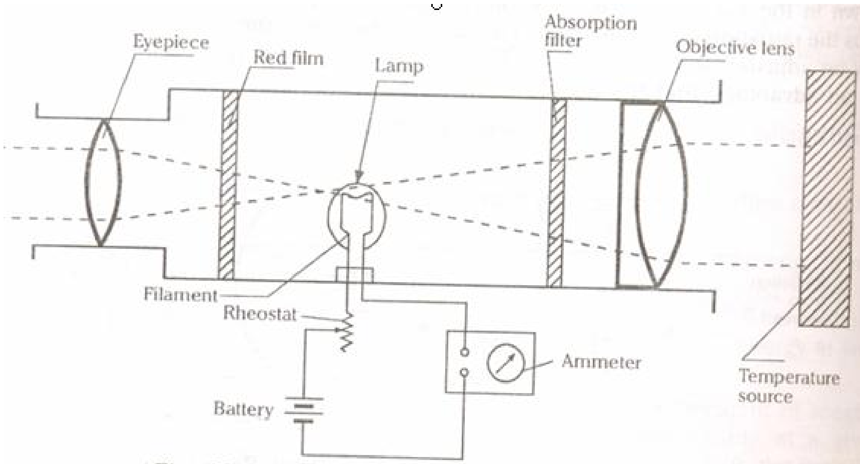


Fig. 12.13 Schematic of an Optical Pyrometer

The eyepiece is then adjusted such that the filament and the source appear superimposed. The filament may appear either hotter or colder than the unknown source as shown in the Fig. The current through the filament is adjusted by means of rheostat.

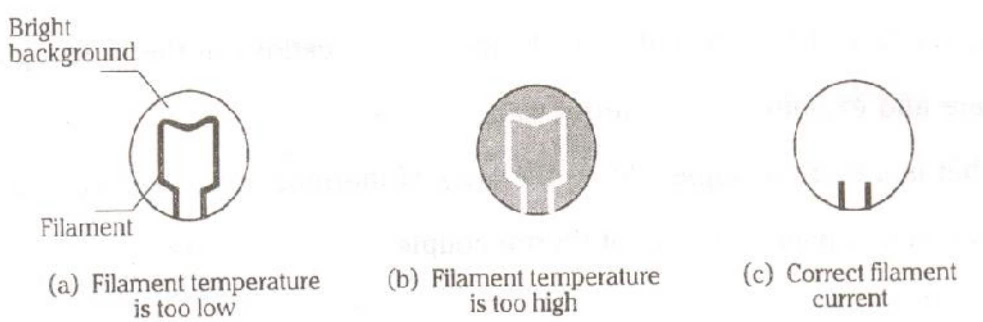


Fig Filament Appearance

When the current passing through the filament is too low, the filament will emit radiation of lesser intensity than that of the source, it will thus appear dark against a bright background as in Fig. (a). When the current is too high it will appear brighter than the background as in Fig. l2.14(b). But when correct current is passed through the filament. The filament "disappears" into the background as in Fig. because it is radiating at the some intensity as the source. In this way the current indicated by the ammeter which disappears the filament may be used as the measure of temperature. The purpose of the red filter is to obtain approximately monochromatic conditions, while an absorption filter is used so that the filament may be operated at reduced intensity.

Strain Measurements

When a system of forces or loads act on a body, it undergoes some deformation. This deformation per unit length is known as **unit strain** or simply a strain mathematically

Strain € = δl /l where, δl = change in length of the body

l= original length of the body.

If a net change in dimension is required, then the term, **total strain** will be used.

Since the strain applied to most engineering materials are very small they are expressed in **"micro train"**

Strain is the quantity used for finding the stress at any point. For measuring the strain, it is the usual practice to make measurements over shortest possible gauge lengths. This is because, the measurement of a change in given length does not give the strain at any fixed point but rather gives the average value over the length. The strain at various points might be different depending upon the strain gradient along the gauge length, then the average strain will be the point strain at the middle point of the gauge length. Since, the change in length over a small gauge length is very small, a high magnification system is required and based upon this, the strain gauges are classified as follows:

i) Mechanical strain gauges
ii) Optical strain gauges
iii) Electrical strain gauges

Mechanical Strain Gauges

This type of strain gauges involves mechanical means for magnification. Extensometer employing compound levers having high magnifications were used. Fig. shows a simple mechanical strain gauge. It consists of two gauge points which will be seated on the specimen whose strain is to be measured. One gauge point is fixed while the second gauge paint is connected to a magnifying lever. which in turn gives the input to a dial indicator. The lever magnifies the displacement and is indicated directly on the calibrated dial indicator. This displacement is used to calculate the strain value.

The most commonly used mechanical strain gauges are Berry-type and Huggen berger type. The Berry extensometer as shown in the Fig. is used for structural applications in civil engineering for long gauge lengths of up to 200 mm.

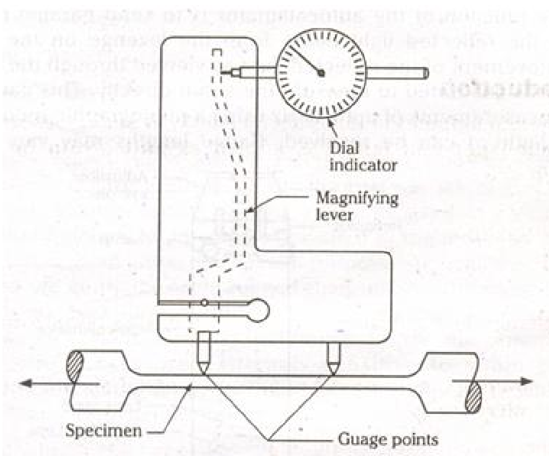


Fig. Mechanical Strain Gauge (Berry Extensometer)

Advantages

1. It has a self contained magnification system.

2. No auxiliary equipment is needed as in the case of electrical strain gauges.

Disadvantages

1. Limited only to static tests.
2. The high inertia of the gauge makes it unsuitable for dynamic measurements and varying strains.
3. The response of the system is slow and also there is no method of recording the readings automatically.
4. There should be sufficient surface area on the test specimen and clearance above it in order to accommodate the gauge together with its mountings

Optical Strain Gauges

The most commonly used optical strain gauge was developed by Tuckerman as shown in the figure It combines mechanical and optical system consisting of an extensometer and an autocollimator. The nominal length of the gauge is the distance from a knife edge to the point of contact of the lozenge. The lozenge acts like a mirror. The distance between the fixed knife edge and lozenge changes, due to loading. Then, the lozenge rotates and if any light beam is falling on it, will be deflected. The function of the auto callimator is to send parallel rays of light and receive back the reflected light beam from the lozenge on the optical system. The relative movement of the reflected light as viewed through the eye-piece of the auto collimator is calibrated to measure the strain directly. This gauge can be used for dynamic measurements of up to 40 Hz using a photographic recorder, and strains as small as 2 µm/m can be resolved. Gauge lengths may vary from 6 mm to 250 mm.

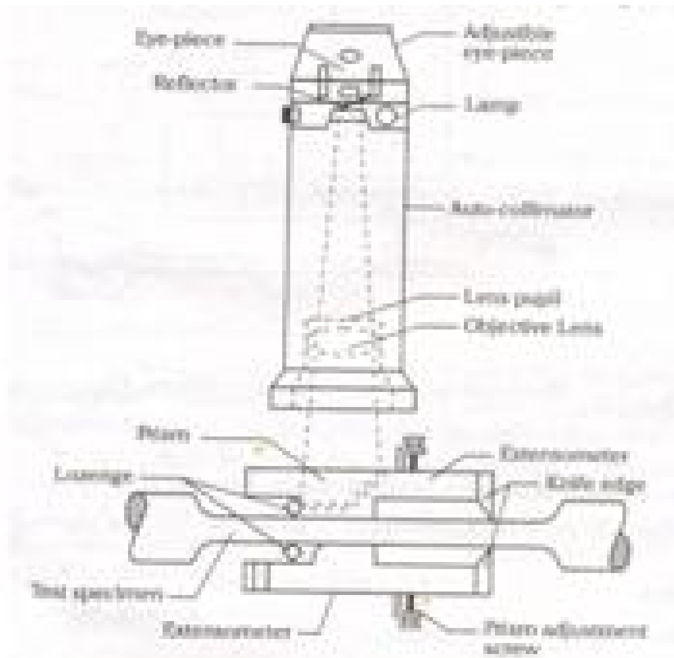


Fig. Tuckerman Optical Extensometer

Advantages:
1. The position of autocolllimator need not be fixed relative to the extensometer, and reading can be taken by holding the autocollimator in hand.

Disadvantages :
1. Limited only for static measurements.
2. Large gauge lengths are required.
3. Cannot be used where large strain gradients are encountered

Electrical Strain Gauges

In electrical strain gauges, a change in strain produces a change in some electrical characteristic. This electrical output can be magnified by some electronic equipment. The electrical methods of measuring strain possess the advantage of high sensitivity and ability to respond to dynamic strains. The electrical strain gauges are classified according to the particular electrical property affected by straining as follows:

(a) Capacitance gauges
(b) Inductance gauges
(c) Piezoelectric gauges
(d) Resistance gauges

The resistance type gauges are by far the most 1 popular, and they have advantages, primarily of size and mass over the other types of electrical gauges. On the other hand, strain-sensitive gauging elements used in calibrated devices for measuring other mechanical quantities are often of the inductive type, whereas the capacitive kind is used more for special-purpose applications. Inductive and capacitive gauges are generally more rugged than resistive one and better able to maintain calibration over long period of time. Inductive gauges are used for permanent installations, such as rolling mill frames for the measurement of rolling loads. Piezoelectric gauges are extremely sensitive to strain but have the disadvantages of non-linearity and relatively high temperature sensitive. They are mainly used for studying dynamic inputs.

Electrical Resistance Strain Gauges

Electrical resistance strain gauges are very widely used for strain measurement. Its operation is based on the principle that the electrical resistance of a conductor changes when it is subjected to a mechanical deformation. Typically an electric conductor is bonded to the specimen with an insulating cement under no-load conditions. A load is then applied, which produces a deformation in both the specimen and the resistance element. This deformation is indicated through a measurement of the change in resistance of an element as follows:

Resistance of a conductor, R= p L/A
where L = Length of the electrical conductor
A = Cross sectional area of the conductor
p = Resistivity of the conductor material.

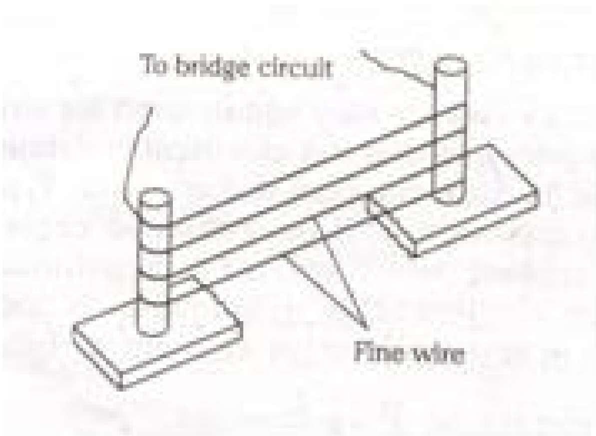
In other words when the conductor is stretched, its length *(*L*)* will increase, area *(*A*)* will decrease and hence the resistance *(*R*)* will increase. Similarly, when the conductor is compressed resistance *(*R*)* will decrease.

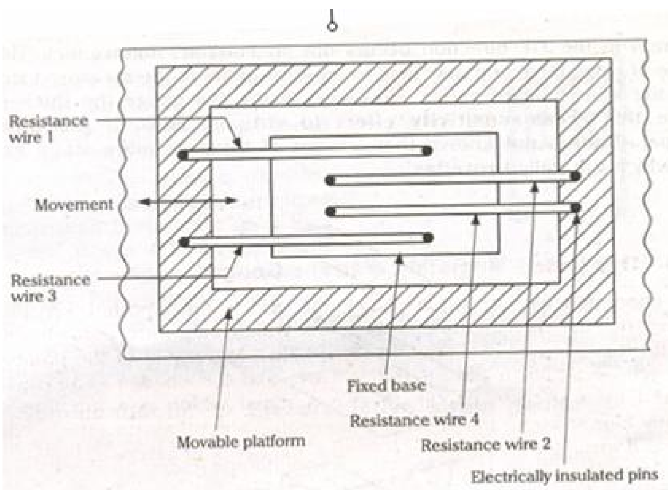
The electrical resistance strain gauges are classified as follows

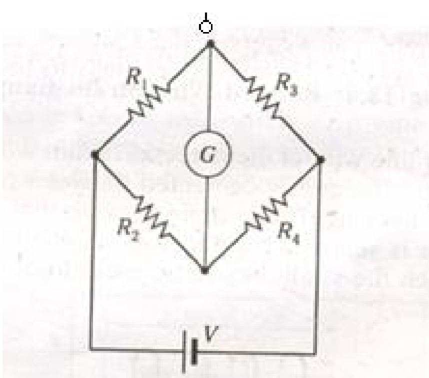
1. Unbonded type strain gauge,
2. Bonded type strain gauge
   a) Wire type b) Foil type.
3. Semi conductor or Piezoresistive strain gauge.

Un bonded Resistance-strain Gauge

In un bonded type of strain gauge, the gird is unsupported. A fine wire is stretched out between two or more points which may form part of a rigid base which itself is strained as shown in Fig. Movement of the points A and B causes tensile strain in the resistance wire, and the change in its resistance is measured by a suitable circuit, to give a signal which is a function of strain. Generally four separate filaments are connected electrically to form a wheatstone bridge and arranged mechanically. so that filaments in adjacent bridge arms are subjected to strains of opposite sign as shown in Fig. (b) and (c). The resistance filaments under tension, are wrapped around electrically insulated pins. When movable platform is moved relative to fixed base, filament tensions are either increased or decreased and the corresponding resistance changes can be calibrated in term of strain. The assembly must provide a built-in prestrain in the grids greater than the maximum compressive strain to be sensed







Bonded Resistance-strain Gauge

The bonded gauge is a suitably shaped piece of resistance metal which is bonded close to the surface whose strain is to be measured. The exploded view Fig. shows a thin wire shaped into a grid pattern, which is cemented between thin sheets of insulating material such as paper or plastic. The grid can also be made from thin metal foil. The assembled gauge is bonded to the surface with a thin layer of adhesive and finally waterproofed with a layer of wax or lacquer. The grid experiences the same strain as the material to which it is bonded. The gauge is most sensitive to the stain along its axial direction *X-X* while the strain in the *Y-Y* direction occurs due to Poisson's ratio effect. This causes change of resistance, and may lead to an error of 2% in the measured strain when using the wire bonded gauge. However, in the foil gauge the thickened ends reduce this cross-sensitivity effect to virtually zero. If the direction of principal strain is not known then cluster of three or more strain gauges are used ,which are called **rosettes.**

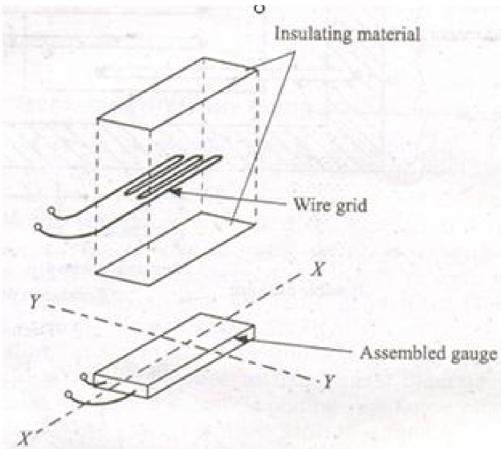


Fig.Bonded Wire strain Gauge

a). Wire type

It consists of a very fine wire of diameter 0.025 mm wounded into a grid shape as shown in fig 13.5 and the grid is cemented between two pieces of thin paper with plastic or ceramic backing. This is done in order that the grid of the wire may be easily handled. This is securely bonded with a suitable cement to the surface of the member in which the strain has to be measured

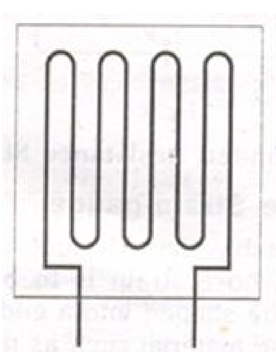
.

Fig. Wire Type Resistance-strain Gauge

b). Foil type resistance strain gauge

These gauges usually employs a foil less than 0.005 mm thick. The common form of foil type gauge consists of a metal foil grid element on a thin epoxy support. Epoxy filled with fiber glass is used for high temperatures. These foil type gauges are manufactured by printing on a thin sheet of metal alloy with an acid-resistant ink, and then the unprinted portion is etched away. Foil gauges have the advantages of improved hysteresis, better fatigue life and lateral strain sensitivity. It is thinner and more flexible, thus permitting it to be applied to fillets and sharply curved surfaces. The common wire or foil gauges are called metallic gauges.

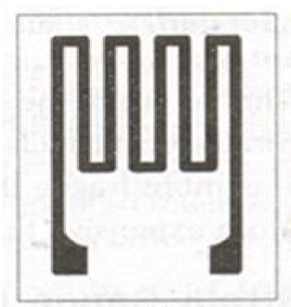


Fig. Single Element. Foil Type Resistance Strain Gauge

Semiconductor or Piezoresistive strain gauge

Semiconductor gauges are cut from single crystals of silicon or germanium in which are combined exact amounts of impurities such as boron which impart certain desirable characteristics. The same types of backing, bonding materials, and mounting techniques as those used for metallic gauges can be used for semiconductor gauges. When the gauge is bonded to a member which is strained, causes changes of current in the semi conductor material. The advantages of semiconductor gauges is their high strain sensitivity which allows very small strains to be measured accurately. A gauge whose electrical resistance increases in response to tensile strain is known as **positive or p-type** semiconductor gauge. On the other hand when the resistance decreases in response to tensile strain then it is known as **negative or n-type** semiconductor gauge.

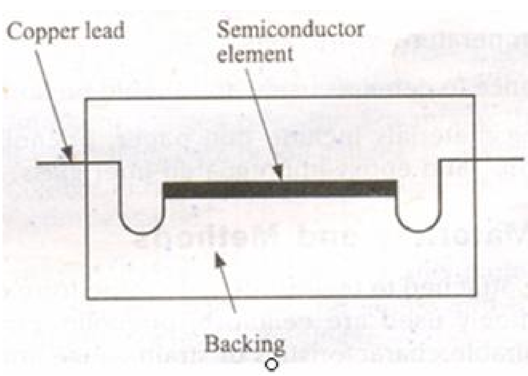


Fig. Semiconductor Strain Gauge

The semiconductor gauge consists of a rectangular filament of about 0.05 mm thick by 0.25 mm wide and gauge length varies from 1.5 to 12 mm as shown in Fig. They are made as thin as possible as the breaking stress of the material rises the cross sectional area decreases and also the gauge can be bent to much smaller radius of curvature without fracture. In addition these gauges have very high temperature coefficients of resistance. The disadvantages of semiconductor gauges are:

I. The output of the gauge is nonlinear with strain
2. Strain-sensitivity is dependent on temperature.
3. It is more fragile than the corresponding wire or foil element.
4. More expensive than ordinary metallic gauges

Strain Gauge Metals

The most common metals used for the manufacture of metallic strain gauges are: alloys of copper and nickel or, alloy of nickel, chromium and iron with other elements in small percentages. Gauges with resistances varying from 60 Ω to 5000 Ω are available. The current carried by the gauges for long periods is around 25 mA to 50 mA.

Strain Gauge Backing Materials

The strain-gauges are normally supported on some form of **backing** material. This provides not only the necessary electrical insulation between the grid and the tested material, but also a convenient carriage for handling the unmounted gauge. Certain types of gauges intended for high temperature applications use a **temporary** backing which will be removed when the grid is mounted. In this case, at the time of installation the grid is embedded in a special ceramic material that provides the necessary electrical insulation and high-temperature adhesion.

The desirable characteristics for backing materials are as follows :
1. Minimum thickness consistent with other factors
2. High mechanical and dielectric strength
3. Minimum temperature restrictions
4. Good adherence to cements used, and should be non-hygroscopic in nature.

Common backing materials include thin paper, phenolic-impregnated paper, epoxy-type plastic films, and epoxy-impregnated fiber glass.

Bonding Materials and Methods

Strain gauges are attached to the test item by some form of cement or adhesive. The adhesives commonly used are cellulose, phenolic, epoxy, cyanoacrylate, or ceramic etc. The desirable characteristics of strain-gauge adhesives are as follows:

1. High mechanical and dielectric strength
2. High creep resistance
3. Minimum temperature restrictions and moisture absorption
4. Good adherence with ease of application
5. The capacity to set fast

Treatment Regarding Preparation and Mounting of Strain Gauges

The following steps should be stricltly followed for proper mounting of the strain gauges:

1. The surface on which the strain gauge has to be mounted must be properly cleaned by an emery cloth and bare base material must be exposed.
2. Various traces of grease or oil etc must be removed by using solvent like acetone.
3. The surface of the strain gauge coming in contact with the test item should also be free from grease etc.
4. Sufficient quantity of cement is applied to the cleaned surface and the cleaned gauge is then simply placed on it. Care should be taken to see that there should not be any air bubble in between the gauge and the surface. The pressure applied should not be heavy so that the cement may puncture the paper and short toe grid

5. The gauges are then allowed to set for at least eight or ten hours before using it. If possible a slight weight may be placed by keeping a sponge or rubber on the gauge.

6. After the cement is fully cured, the electrical, continuity of grid must be checked by ohm meter and the electrical leads may be welded.

Problems Associated with Strain-gauge Installations

The problems associated with strain-gauge installations generally fall into three categories.

1. **Temperature effects**: Temperature problems arise because of differential thermal expansion between the resistance element and the material to which it is bonded. Semiconductor gauges offer the advantage that they have lower expansion coefficient than either wire or foil gauges. In addition to the expansion problem, there is a change in resistance of the gauge with temperature, which must be adequately compensated

2. **Moisture absorption**: Moisture absorption by the paper and cement can change the electrical resistance between the gauge and the ground potential and thus affect the output resistance readings.

3. **Wiring Problems** : This problem arise because of faulty connections between the gauge-resistance element and the external read out circuit. These problems may develop from poorly soldered connections or from inflexible wiring, which may pull the gauge loose from the test specimen or break the gauge altogether.

Gauge Factor or Strain Sensitivity

The electrical resistance of a metallic conductor varies directly as its resistivity and length and inversely as its cross-sectional area i.e.,

$$R = pL/A$$

The metallic crystal lattice forms a regular atomic pattern and the particular form of bonding between the atoms-metallic bonding-involves mutual sharing of all the valency electrons by all the atoms in the metal. Thus, current passes along a conductor in the form of directional motion of the electrons, called electron flow. The increased length and decreased area of a gauge under tensile strain. Accounts for part of the increase in resistance as the metallic lattice will suffer distortion but these changes alone do not fully account for the total resistance change, thus other changes in the metal lattice must also occur bringing about a change in the resistivity of the metal. This latter effect is an important consideration as it is well known that the resistivity of metals also changes with temperature.

The gauge factor G_F is the ratio of change of resistance *dR/R* to the change of gauge length *dL/L,* thus

$$G_F = \frac{dR/R}{dL/L} \qquad \text{where } R \text{ is the nominal resistance of the gauge.}$$

The gauge factor is supplied by the manufacturer and may range from 1.7 to 4 depending on the length of the gauge. Thus, provided that a means is available for measuring the change of resistance, strain can be determined and hence stress. An increase in gauge factor increases the sensitivity, but it is limited in metal gauges, largely because of the relatively low resistivity of metals, a limitation which is overcome in semiconductor strain gauges which have gauge factors of the order of ± 100 or more

Basic Wheatstone Resistance Bridge and Methods of Strain Measurement

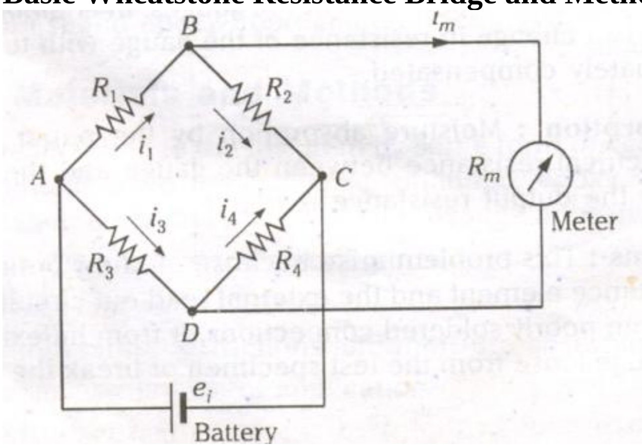


Fig. Basic Wheatstone Bridge Circuit

Electrical resistance type of strain gauges uses highly sensitive wheatstone bridge circuit for the measurement of strain. It consists of four resistance arms with a source of energy (battery) and a detector (meter) as shown in Fig. The basic principle of the bridge may be applied in two different ways the **null method** and the **deflection method.** Let us assume that the resistances have been adjusted so that the bridge is balanced (i.e $e_{BD} = 0$). In order for the bridge to balance, the ratio of resistances of any two adjacent arms must be equal to the ratio of resistances of the remaining two arms, taken in the same sense (i.e., $R_1 / R_3 = R_2 / R_4$). Suppose if anyone of the resistance say R_1 changes, it will unbalance the bridge, and a voltage will appear across *BD* causing a meter reading. The meter reading is an indication of the change in R_1 and actually can be utilized to compute this change. This method of measuring the resistance change is called the **deflection method,** since the meter deflection indicates the resistance change.

In the **null method,** one of the resistors is adjusted manually. Thus, if R_1 changes, causing a meter deflection, R_2 can be adjusted manually until its effect just cancels that of R_1 and the bridge is returned to its balanced condition.

The adjustment of R_2 is guided by the meter reading, R_2 is adjusted so that the meter returns to its null or zero position. In this case the numerical value of the change in R_1 is related directly to the change in R_2 required to effect balance.

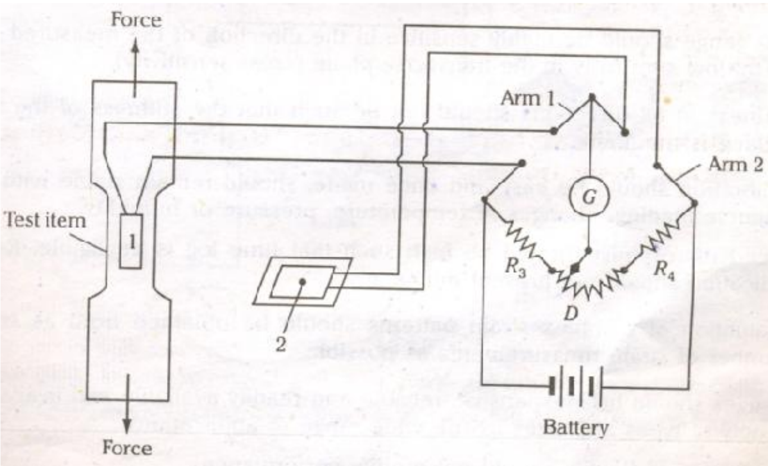


Fig. Wheatston Bridge Arrangement for strain Measurement

A wheatston resistance-bridge arrangement is particularly convenient for use with strain gauges because it may be easily adjusted to a null for zero strain, and it provides means for effectively reducing or eliminating the temperature effects. The Fig. shows a simple resistance bridge arrangement for strain measurement. In which an arm 1 consists of a strain-sensitive gauge mounted on the test item. Arm 2 is formed by a similar gauge mounted on a piece of unstrained material as that of the test item and placed near the test location so that the temperature will be same. Arms 3 and 4 contains fixed resistors selected for good stability and portions of slide-wire resistance, *D* is required for balancing the bridge.

In the deflection method of strain measurement continuous reading from one change of strain to another without rebalancing. This is convenient but less accurate as the bridge output tends to non-linearity if the strain is successively increased or decreased.

In the null balance method after each successive change in strain, the bridge is rebalanced. As the strain indicator will be calibrated under null balance conditions this method offers the highest accuracy while maintaining linear bridge output.

Requirements for Accurate Strain Measurement

Ideally the gauges used for strain measurement should conform to the following requirements :

1. The gauge should be small in size and easy to mount on the component.
2. The profile should be as low as possible so that it will respond in close agreement with the changes in the surface to which it is fixed.
3. The gauge should be highly sensitive in the direction of the measured strain but of low sensitivity in the transverse plane (cross-sensitivity).
4. Stiffness in all directions should not be such that the stiffness of the tested surface is modified.
5. Calibration should be easy and once made, should remain stable with time, dynamic loading, changes of temperature, pressure or humidity.
6. Speed of response should be high such that time log is negligible. Remote indication should not present difficulties.
7. Evaluation of complex strain patterns should be obtained from as small a number of strain measurements as possible.
8. Gauges should be inexpensive, reliable and readily available and available in variety of types and sizes to suit wide range of applications.
9. Immersion in liquids should not modify performance.

A bonded resistance strain gauges satisfy many of these requirements.

Temperature Compensation

Resistance type strain gauges are sensitive to temperature changes. Any variation in the temperature influences the strain gauge readings as follows :

1. The gauge factor of the strain gauge is affected by temperature owing to creep.
2. The resistance of the strain gauge element varies with a change in the temperature.
3. Strain may be induced in the gauge due to the differential expansion between· the test member and the strain gauge bonding material.

These temperature effects may be compensated by

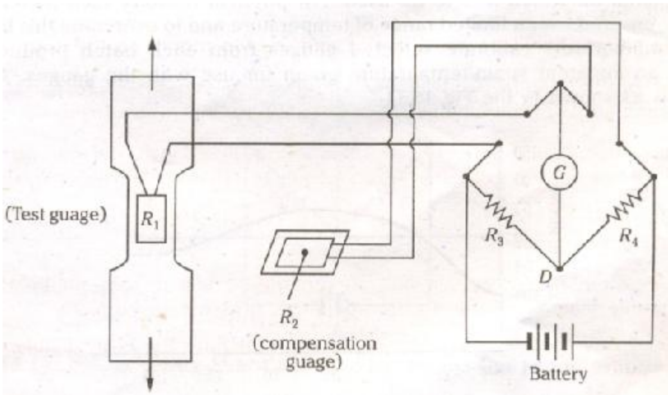
a) Adjacent - arm compensating gauge
b) Self - temperature compensation

a). Adjacent - Arm compensating gauge

Consider the bridge arrangement for strain measurement as shown in the

Fig. The condition for the bridge balance is given by

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$



If the strain gauges in arms 1 and 2 are alike and mounted on similar materials, and if both gauges experience the same resistance shift ΔRt, caused by temperature change, then

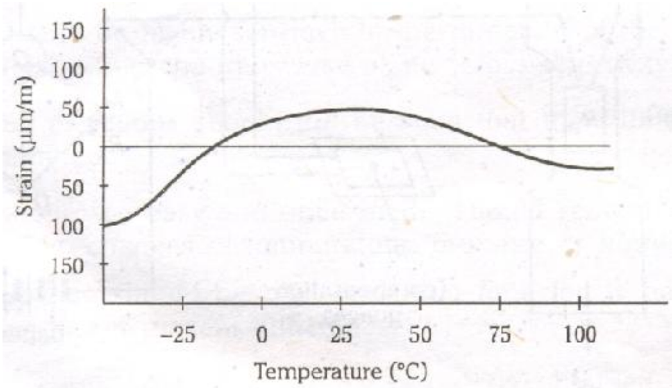
$$\frac{R_1+\Delta R_t}{R_2+\Delta R_t}=\frac{R_3}{R_4}$$

It is clear that any changes in the resistance of gauge 1 due to temperature is cancelled by similar changes in the resistance of gauge 2 and the bridge remains in balance and the output is unaffected by the change in temperature.

(b) Self-temperature Compensation

A completely self-compensated gauge should exhibit zero change of resistance with change of temperature when bonded to a specified material. Consider a conventional advance gauge bonded to a steel base. The coefficient of linear expansion of this alloy is 15 μm^o c/m. while the coefficient of steel is 12 µmo c/m; thus under rising temperature the free expansion of the gauge is prevented and as a result the gauge is in compression and dR/R from this source is negative. Conversely, the thermal coefficient of resistance of the gauge itself is positive, thus dR/R from this source is positive. If these resistance changes are equal and opposite then change of resistance due to temperature is zero from these sources. It is of interest to note that a similar gauge bonded to aluminum, coefficient 12 µmo c/m, would be in tension. The most widely used self-compensated gauges are made from specially prepared "selected melt" alloys for use on specified materials.

The temperature coefficient of resistance of the gauge is, by suitable heat treatment during manufacture, matched to the coefficient of linear expansion of the material on which it is to be used. For physical reasons such matching. can only be ensured over a limited range of temperature and to overcome this limitation the manufacturers calibrate selected gauges from each batch produced and supply an apparent strain/temperature graph for use with the gauges. A typical graph is as shown in the Fig

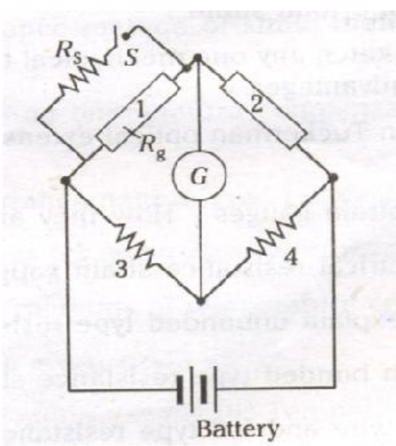


Another type is made by connecting two different wires in series to form the grid, one having a positive and the other a negative gauge factor. Increase in resistance in one wire due to temperature is almost equally compensated by a similar decrease in the other. Such gauges must still be matched for use on specified materials.

Calibration of Strain Gauges

Calibration refers to a situation where an accurately known input is applied to the system and the corresponding output is measured. This procedure cannot be applied in bonded resistance strain gauges because of its nature. Normally, the gauge is bonded to the test item to measure the strains. Once the gauge is bonded it cannot be transferred to a known strain situation for calibration. In the calibration of strain gauges the value of both the gauge factor and gauge resistance should be know accurately.

The gauge factor for each of the gauge is provided by the manufacturer within an indicated tolerance of about ± 0.2%. The method consists of determining the systems response by introducing a known small resistance change at the gauge and calculating the equivalent strain. The resistance change is introduced by shunting a relatively high-value precision resistance across the gauge as shown in the Fig. When switch S is closed, the resistance of the bridge arm 1 is changed by a small amount ΔR. If R_g is the gauge resistance, R_s is the shunt resistance and the resistance of arm 1 after the switch S is closed becomes.



$$\frac{R_g R_s}{R_g + R_s}$$

Fig. Bridge using Shunt Resistor for Strain Gauge Calibration

The resistance of arm 1 before the switch S is closed = R_g

The resistance of arm 1 after the switch S is closed = $\dfrac{R_g R_s}{R_g + R_s}$

$$\therefore \; Change \; in \; resis\tan ce \;\; \Delta R = \frac{R_g \, R_s}{(R_g - R_s)} - R_g \; = \frac{-R_g^{\;2}}{R_g \; = R_s}$$

$$The \;\; strain \; in \; the \; gauge \;\; £ \; = \frac{1}{G_F}\left(\frac{R_g}{R_g \; + R_{\;s}}\right)$$

By substituting Δ R for Δ Rg, the equivalent strain is found to be

$$£_e = \frac{-1}{G_F}\left(\frac{R_g}{R_g + R_s}\right) \qquad \text{--------(1)}$$

A number of equivalent strain values can be obtained by shunting different shunt resistors across R_g A graph between resistance change ΔR and the equivalent strain was plotted. The strain corresponding to the measured resistance changes when the gauge is bonded on the test member can be read from the €e Vs ΔR plot.